Virtual temperature sensors for second-life battery packs

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Abstract

To reduce the costs of repurposing used electric vehicle batteries for electricity grid energy storage, second-life battery packs can be operated in their original housings without any modifications. One of the drawbacks of this approach is the necessity to rely on the original manufacturer's placement of temperature sensors, which may not meet the temperature monitoring requirements of a given second-life project developer. This study proposes a novel 'virtual temperature sensor' methodology as a means of predicting real-time temperatures at unmonitored locations within a sealed battery pack based on measurements from the original temperature sensors. A proof-of-concept is demonstrated for a 2019 Tesla Model 3 SR+ battery pack, showing that a simplified thermal model can be trained with empirical data to accurately predict an unmonitored temperature over a wide range of operating conditions. A generalized methodology is also described to guide the development of similar models for other second-life battery packs. This non-invasive technology can de deployed by the second-life battery industry to enhance safety, performance, and utilization of used electric vehicle battery packs without the need for costly modifications.

Keywords
Electric vehicle
Second life
Battery
Temperature
Thermal model

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Symbols:

$A_{x\infty}$	m^2	Area of heat dissipation from cell X to ambient environment
$A_{z\infty}$	m^2	Area of heat dissipation from cell Z to ambient environment
$A_{z'x}$	m^2	Area of lateral heat diffusion from cell Z to cell X
$A_{z'z}$	m^2	Area of axial heat diffusion from cell Z core to cell Z upper surface
c_p	J/kg · K	Specific heat capacity
Err_x	°C	Model error for temperature of cell X
Err_z	°C	Model error for temperature of cell Z upper surface
Err_{VTS}	°C	Total model error
$h^*_{conv,x}$	W/m^2 · K	Convection coefficient reference value for cell X
$h^*_{conv,z}$	W/m^2 · K	Convection coefficient reference value for cell Z
I	A	Average cell current
i	-	Data interval number
$k_{z'x}$	W/m	Lateral thermal conductivity between cell Z and cell X
$k_{z'z}$	W/m	Axial thermal conductivity within cell Z
$L_{z'x}$	m	Length of lateral heat diffusion from cell Z to cell X
$L_{z'z}$	m	Length of axial heat diffusion from cell Z core to cell Z upper surface
m_{cell}	kg	Mass of single cell
m_x	kg	Mass of cell X and adjacent liquid channel
m_z	kg	Mass of cell Z upper surface
$m_{z'}$	kg	Mass of cell Z core
$\dot{q}_{gen}$	W	Heat generation rate
q^{cyc}_{gen}	J	Heat generation for a complete discharge-charge cycle
t	s	Time
T_∞	°C	Temperature of ambient environment
T_x	°C	Temperature of cell X exposed cylindrical wall surface
T_z	°C	Temperature of cell Z exposed upper surface
$T_{z'}$	°C	Temperature of cell Z core
ΔT^{chg}_{ad}	°C	Theoretical adiabatic cell temperature change during charge
ΔT^{dis}_{ad}	°C	Theoretical adiabatic cell temperature change during discharge
V_x	V	Voltage of cell X
V_z	V	Voltage of cell Z
α_Θ	-	Temperature generation rate correction factor
ε_{rad}	-	Radiative emissivity
Θ_x	°C/s	Temperature generation rate of cell X
$\Theta_{z'}$	°C/s	Temperature generation rate of cell Z core
$\tau_{x\infty}$	s	Time variable of heat dissipation from cell X to ambient environment
$\tau_{z\infty}$	s	Time variable of heat dissipation from cell Z to ambient environment
$\tau^*_{x\infty}$	s	Constant reference value for $\tau_{x\infty}$
$\tau^*_{z\infty}$	s	Constant reference value for $\tau_{z\infty}$
$\tau_{z'x}$	s	Time constant of lateral heat diffusion from cell Z to cell X
$\tau_{z'z}$	s	Time constant of axial heat diffusion from cell Z core to cell Z upper surface
τ_x	s	Time variable of overall heat transfer through cell X
τ_z	s	Time variable of overall heat transfer through cell Z

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λ_x	-	Ratio of m_x to $m_{z'}$
λ_z	-	Ratio of $m_{z'}$ to m_z

1. Introduction

Second-life batteries from electric vehicles (EVs) can help meet the increasing global demand for affordable electricity grid energy storage. Once an EV battery pack is removed from the vehicle, it can be repurposed for electricity grid applications through one of two main methods: (1) the pack is opened and disassembled into modules or cells which are then repackaged into a different battery housing; or (2) the pack is repurposed in its original housing. The disassembly and repackaging option offers the opportunity to select preferred modules/cells and customize the enclosure, sensors, circuitry, and thermal management of the second-life battery system. However, these processing measures may require significant time, labour, and materials, which is economically disadvantageous.

Since EV battery packs are already designed with built-in sensors, battery management system (BMS), and protection circuitry, it is possible to repurpose an EV battery pack 'as-received' by simply connecting to the main power port and the built-in BMS without ever opening or modifying the pack. This option has clear economic advantages, but there is little opportunity to customize the pack for second-life service. One potential disadvantage is temperature monitoring, since a second-life battery developer will only be able to monitor the built-in temperature sensors installed at specific locations within the pack by the original manufacturer.

In some cases, the original temperature sensor locations may not adequately capture the thermal performance metrics which the second-life energy storage developer wishes to monitor. This is especially likely if the EV battery will be operated with a different thermal management system in second life compared to EV service. For example, many modern EV batteries are designed with liquid-cooling capability in EV service, but the built-in temperature monitoring system may not be adequate if a second-life project developer elects to operate the batteries with passive cooling. As an alternative to costly retrofitting measures, numerical models could be used to accurately predict temperatures at desired locations in the pack without adding physical sensors.

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Battery temperatures can be modeled if the heat generation rates, heat transfer rates, and thermal properties are known or can be estimated. While it is sometimes acceptable to employ simplified thermal models featuring a single lumped heat transfer coefficient to capture the combined effects of conduction, convection, and radiation [1], [2], most studies on battery modules and packs employ more advanced models to account for temperature distributions throughout the battery. Three-dimensional heat conduction problems can be solved numerically by applying finite element analysis with computational software tools [3], [4], [5], which require initial conditions as well as boundary conditions for convection and/or radiation at the outer surfaces of the battery. Many such models have been described in the literature, with simulations potentially requiring many hours depending on the size of the battery, the number of nodes used in the model, and the degree of simplification applied [6]. The relatively greater size and complexity of EV battery packs are generally associated with greater computational requirements.

Battery thermal models generally employ theoretical or empirical approaches to describing the various thermal properties and processes of the battery. Theoretical models require detailed specifications of the battery design in order to calculate the variable rates of heat generation and heat transfer based on electrochemical and thermal processes. Required specifications may include electrode thickness and porosity, electrolyte conductivity and diffusivity, and particle radius [1], [7], [8]. While these specifications will be known by the battery manufacturer, most second-life battery developers will not have access to this proprietary information, making theoretical thermal models unsuitable for most second-life applications. Relatively simple expressions for the irreversible and reversible heat generation rates have been widely adopted [9], with the only required input parameters being the cell equilibrium voltage and entropic heat coefficient as functions of cell state-of-charge (SOC). While it is possible to determine these parameters experimentally for a given battery, the experiments may require several days in a specialized test chamber and may not be consistent over wide ranges in state-of-health (SOH), making this approach impractical for most second-life applications.

Some studies have estimated certain battery model parameters by fitting the simulated results to empirical test data. For example, Ye et al. described an electro-thermal model for which the lithium-ion diffusivity and chemical reaction rate were estimated by fitting the model to experimental current pulse tests [2]. However, their model still required specifications for other

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battery parameters which would be unknown in most second-life applications. Bazinski and Wang used polynomial expressions to estimate the rates of heat generation and heat transfer for a single lithium-ion pouch cell by fitting the model to battery surface temperature measurements obtained from empirical tests [10]. Since their model assumed lumped capacitance for a single pouch cell, it is not directly applicable to EV battery packs where nonuniform temperature distributions must be accounted for. Ramotar et al. designed a thermal equivalent circuit model to empirically characterize the thermal behaviour of a liquid-cooled Chevrolet Volt hybrid EV battery pack, achieving reasonable results with reduced computational requirements compared to finite element analysis [11]. However, their model still required characterization of over 40 individual parameters.

This study proposes a novel 'virtual temperature sensor' (VTS) methodology for numerical modelling of unknown temperatures at desired locations in an EV battery pack based on known temperatures measured by original sensors. To demonstrate a proof-of-concept, a VTS model is developed for a Tesla Model 3 Standard Range Plus (SR+) EV battery pack. A simplified thermal model is developed to establish a robust mathematical relationship between two specific cells in the pack. The proposed model does not require specification of any thermal or electrochemical properties for the battery. Instead, the model is governed by a set of seven constant parameters which must be empirically determined using experimental test data.

The remainder of the article is structured as follows: Section 2 describes the proposed thermal model; Section 3 explains the techniques used for parameter optimization; Section 4 discusses the accuracy and versatility of the proposed model; Section 5 reviews opportunities for future VTS research; Section 6 summarizes the main conclusions of the study. Some of the finer details of the proposed VTS model are provided in the Supplementary Materials, which are referenced throughout the article.

2. Thermal model

As described in Section 1, battery thermal models typically require proprietary knowledge of cell design parameters and/or advanced three-dimensional computational software tools to simulate complex heat generation and heat transfer processes. This section will present a simplified thermal

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model to enable quantification of key heat flows between specific temperature nodes for the purpose of a VTS model.

2.1. Battery characteristics

The EV battery used for this study is the 2019 Tesla Model 3 SR+ battery pack pictured in Figure 1. This EV battery pack features four identical modules each containing 24 series-connected cell groups of 31 parallel-connected cylindrical cells. The cells are of 2170 format (21 mm diameter and 70 mm height) with a lithium nickel cobalt aluminum (NCA) positive electrode and a graphite-silicon blended negative electrode [12]. The pack is nominally rated for 50 kWh of discharge energy, but at the time of testing the pack was found to have a 4 h discharge energy of 43.4 kWh. This corresponds to a state-of-health (SOH) of approximately 87%.

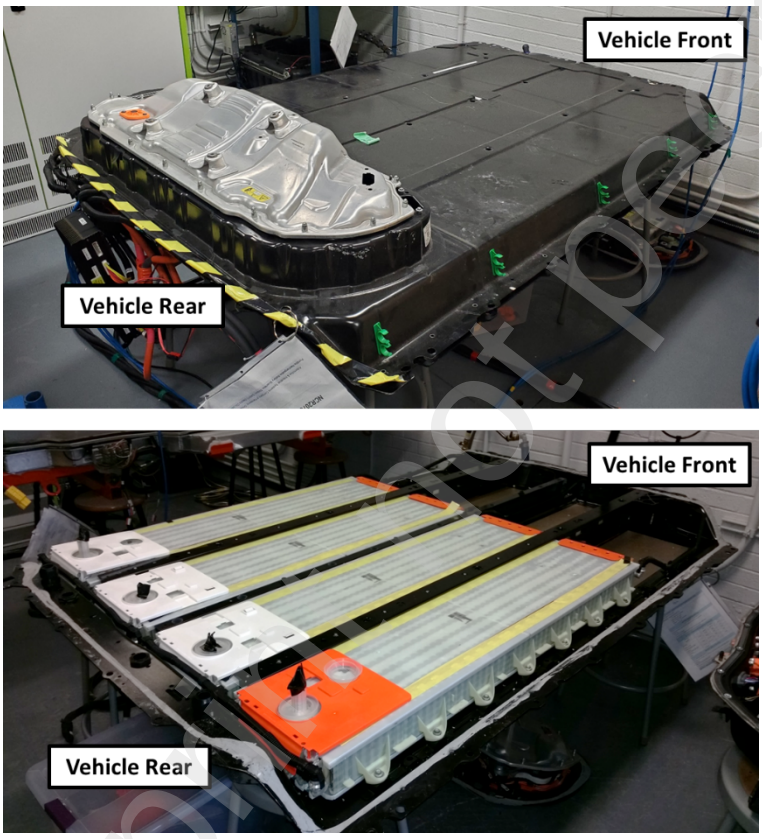


Figure 1 2019 Tesla Model 3 SR+ battery pack with housing cover on (top) and off (bottom), showing four distinct modules.

This battery pack is designed to be liquid-cooled in EV service. The four modules each feature seven parallel liquid channels that pass between cylindrical cell walls in a corrugated fashion along the length of the modules. Liquid flows from the front of the vehicle towards the rear, as shown in

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Figure 2. The figure also shows the built-in temperature monitoring system, consisting of a pair of thermistors placed towards the rear of each module where the liquid exits. These sensor locations are appropriate for capturing maximum pack temperatures when the battery is liquid-cooled, but some second-life energy storage developers may elect to operate the battery pack with passive cooling for the sake of simplicity and cost. In this case, Tesla's built-in temperature monitoring system would not be able to capture the highest pack temperature T_{high} (likely occurring near the middle of the inner modules), nor the increased pack temperature differential between T_{high} and the lowest pack temperature T_{low} (likely occurring near the outer module corners). This additional information would help ensure safe and economical second-life operation for a passive-cooled system. Specific examples might include more appropriate control of power envelopes or more accurate projection of capacity degradation.

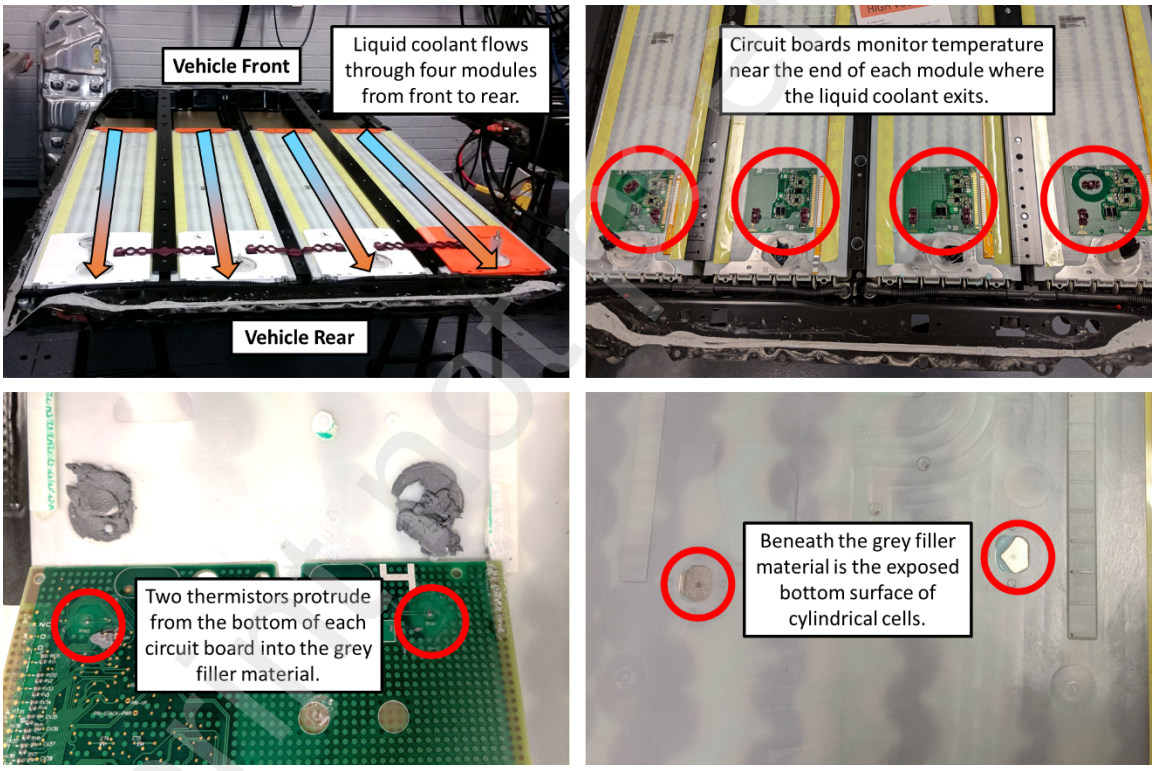


Figure 2 The built-in temperature monitoring system of a Tesla Model 3 SR+ battery pack.

The Tesla Model 3 SR+ battery is a good example of the complexity of heat transfer in EV battery packs. This is illustrated in Figure 3, where the red arrows represent conduction pathways and the blue arrows represent convection and radiation pathways. The liquid channels are indicated in grey, while the green area represents all the cylindrical cells and the filler material that occupies

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the spaces between the cells. When the battery is operating, the generated heat is transferred through and away from the pack in a variety of ways, including the following:

- Horizontal conduction within the modules though the cylindrically-wound cell stacks, cell housings, and filler material.
- Horizontal conduction within the modules along the length of the liquid cooling channels to the liquid manifolds at either end.
- Horizontal conduction within the modules through alternating layers of cells/filler and liquid channels.
- Vertical conduction within the modules through the height of the cells, filler, and liquid channels.
- Natural convection from the modules and liquid manifolds to the internal air.
- Natural convection from the internal air to the inner surface of the housing cover.
- Radiation from the modules and liquid manifolds to the inner surface of the housing cover.
- Conduction from the inner surface of the housing cover and electronics housing to their outer surfaces.
- Natural convection from the outer surface of the housing cover to the external air.
- Conduction from the bottom of the modules to the housing base.
- Conduction from the housing to the mounting/racking assembly.
- Natural convection from the housing base to the external air below.
- Forced convection from the liquid channels to the liquid coolant (only when operated with active-liquid thermal management).
- Radiation from the outer surfaces of the housing to the surroundings.

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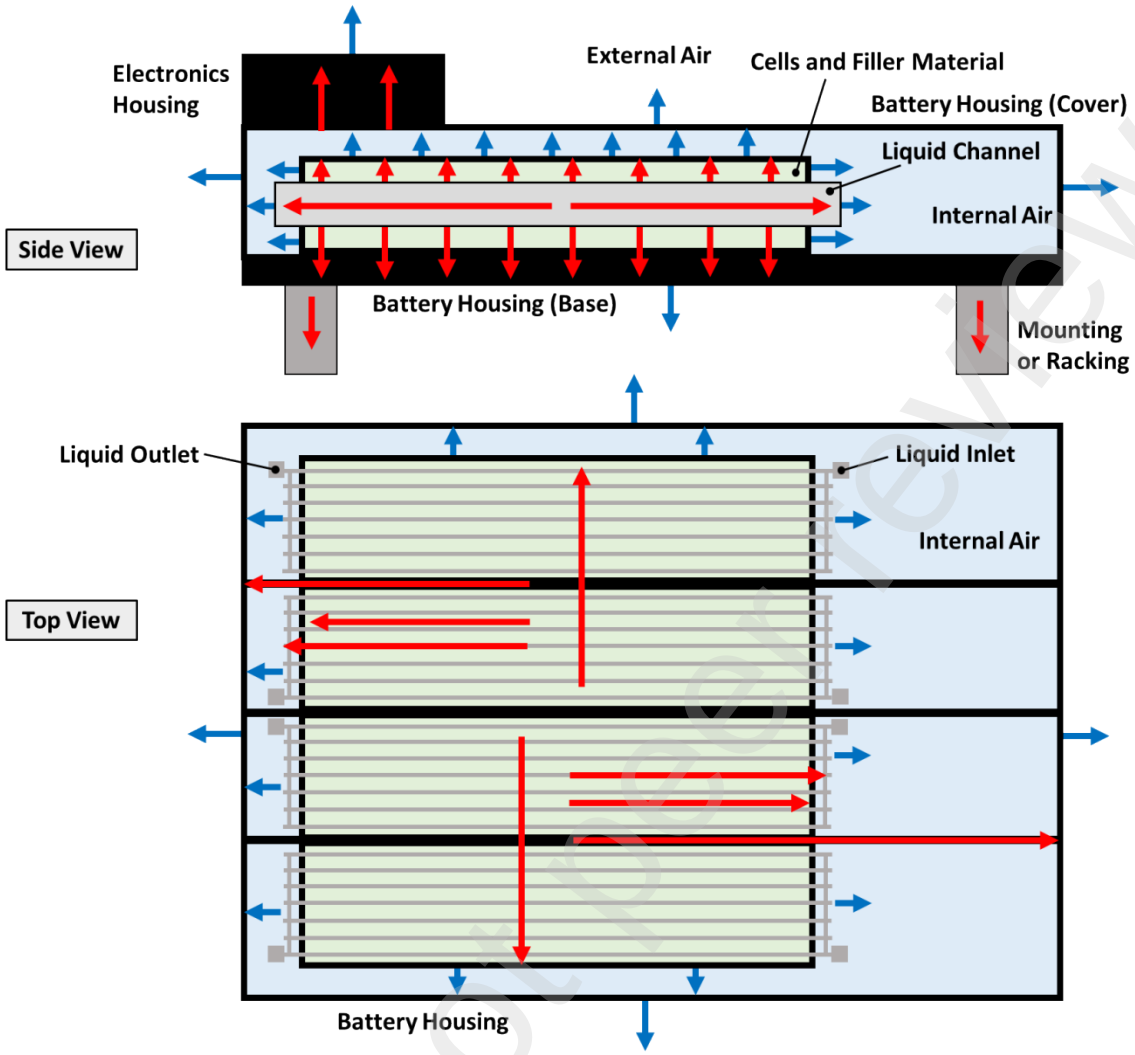


Figure 3 Illustration of heat transfer in an operational Tesla Model 3 SR+ battery pack. Red arrows represent heat conduction pathways. Blue arrows represent convection and radiation pathways. Not drawn to scale.

When the battery is operated with liquid cooling, forced convection likely dominates the thermal system [13]. When the battery is operated with passive cooling, the numerous other heat transfer pathways must be accounted for in the thermal model. Due to the anisotropic heat conductivity of cylindrical cells [14], the modules will share this anisotropy. The presence of the liquid channels means that the module thermal conductivity is anisotropic in three dimensions, since there are essentially 17 series thermal resistances through the width of each module (alternating between seven liquid channels and eight layers of cells/filler, as well as the outer module walls), and two parallel thermal resistances along the length of each module (the liquid channels and the cells/filler). The liquid channels themselves are also anisotropic since there will either be liquid

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coolant or air occupying the space within the channels. The composite thermal conductivity of the cells and filler is also dependent on the properties and geometry of each. The many influential factors of geometry and material properties may be difficult to quantify without manufacturer specifications.

2.2. Simplified model

Given the numerous complexities and sources of uncertainty in developing a complete three-dimensional thermal model, a simplified empirical model is proposed here for the purpose of developing a practical VTS model. Figure 4 illustrates the chosen model, which features only two cells: cell Z representing a cell located in the interior of an outer module, and cell X representing a cell located at the corner of the same outer module. Tesla's built-in temperature monitoring system includes a thermistor applied to the top surface of cell Z. The temperature of cell X is not monitored by Tesla's built-in system, but it is expected be near T_{low} based on the location of cell X. In this study, T_{z} refers to the temperature at the top of cell Z, while T_{x} refers to the temperature at the exposed cylindrical wall of cell X. A temperature sensor can be applied to the exposed cylindrical wall of cell X to collect empirical data for the VTS model. It is more difficult to collect empirical data for cells in the centre of the pack where T_{high} is expected to occur, due to a lack of exposed cell surface. Cell X is therefore chosen as the "unknown" cell for this pilot study.

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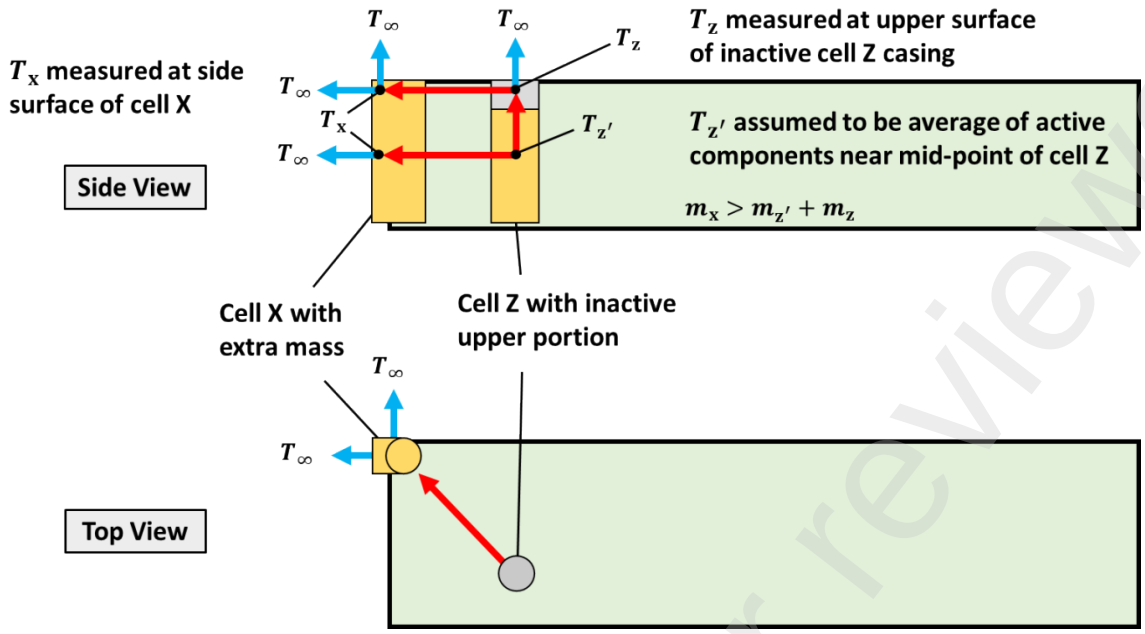


Figure 4 Illustration of the proposed simplified thermal model. Red arrows represent heat conduction pathways. Blue arrows represent convection and radiation pathways. Yellow masses generate heat, while grey masses do not. Not drawn to scale.

As shown in Figure 4, the module is treated as though cell X and cell Z are the only heat generation sources, while the remainder of the module volume is thermally insulating except for discrete thermally conductive pathways. The heat capacity of the internal air and battery housing cover are neglected, allowing for simplified heat dissipation from the exposed cell surfaces directly to the external air. Cell Z is divided into two discrete masses as a simple means of modelling axial heat diffusion within cell Z, the effects of which were observed experimentally (see Section 4.1). The lower mass $m_{z'}$ is modelled as an active cell stack which generates heat and has an average temperature $T_{z'}$ that cannot be measured directly. The upper mass m_z is modelled as an inactive cell housing material which does not generate heat and has a surface temperature of T_z that is measured by a temperature sensor. Note that in the case of the Tesla Model 3 battery, the steel cell housing serves as part of the negative electrode and therefore carries current, but the generated heat will be small compared to the heat generated by the cell stack and is therefore neglected. A heat conduction pathway is placed between $T_{z'}$ and T_z, while a convection/radiation pathway is placed between T_z and the ambient temperature T_∞.

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Unlike cell Z, the effects of heat diffusion within cell X were not observed experimentally, possibly due to a masking effect caused by greater heat dissipation from cell X compared to cell Z. Cell X is therefore treated as a single mass m_x with an average surface temperature T_x that is measured by a temperature sensor. A heat conduction pathway is placed between $T_{z'}$ and T_x as well as between T_z and T_x, while a convection/radiation pathway is placed between T_x and T_∞. In the real system, there is likely lateral heat conduction into cell Z from warmer regions of the pack. This is not shown explicitly in Figure 4. However, the net lateral heat conduction should be away from cell Z, and the simplified model in Figure 4 aims to approximate this net effect.

It is important to note that cell X is located next to the extended end of the liquid channel and the affixed liquid outlet, which provides additional inactive heat capacity at the corner of the module compared to the interior. Since heat diffusion from cell X to this extra mass is not included in the model, this effectively means that the model assumes $m_x > m_{z'} + m_z$. The specific heat capacity is assumed to be uniform.

The model in Figure 4 is a significant simplification of a complex thermal system, but it provides an approximation of several key features that govern the real system:

- Heat conducts axially from the core of cell Z to the upper surface
- Heat conducts laterally away from cell Z to cooler regions near the module edges
- Heat conducts laterally into cell X from warmer regions in the module interior
- Heat dissipates from the exposed cell surfaces to the external environment

The goal of this study is to demonstrate that the proposed thermal model in Figure 4 can be used to accurately predict T_x based on real-time measurements of T_z and T_∞ over a practical range of operating conditions. The metric for accuracy is defined to be within about 0.5 °C since this is a common thermistor measurement resolution.

2.3. System equations

The proposed thermal model in Figure 4 includes three masses subject to heat transfer. The first step in deriving mathematical equations for this model is to apply an energy balance to all three

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masses based on Eq. (1), where c_p is the specific heat capacity, $\dot{q}_{\text{gen}}$ is the heat generation rate, $\dot{q}_{\text{in}}$ is the heat inflow rate, and $\dot{q}_{\text{out}}$ is the heat outflow rate.

$$mc_p\frac{\mathrm{d}T}{\mathrm{d}t} = \dot{q}_{\text{gen}} + \dot{q}_{\text{in}} - \dot{q}_{\text{out}} \tag{1}$$

Th energy balance equations for m_{x}, $m_{z'}$, and m_z are stated in Eq. (2), (3), and (4) respectively. In these equations h, k, A, and L respectively represent heat dissipation coefficient (i.e. convection and radiation combined), thermal conductivity, heat transfer area, and heat transfer length. The subscripts attached to these four variables denote the specific heat transfer mode in question. For example, A_{zx} represents the area through which heat transfers from Z to X from the perspective of Z, while A_{xz} represents the effective heat transfer area from the perspective of X.

$$m_{\text{x}}c_p\frac{\mathrm{d}T_{\text{x}}}{\mathrm{d}t} = \dot{q}_{\text{gen,x}} + \frac{k_{\text{z'x}}A_{\text{z'x}}}{L_{\text{z'x}}}(T_{\text{z'}} - T_{\text{x}}) + \frac{k_{\text{zx}}A_{\text{zx}}}{L_{\text{zx}}}(T_{\text{z}} - T_{\text{x}}) - h_{\text{x}\infty}A_{\text{x}\infty}(T_{\text{x}} - T_{\infty}) \tag{2}$$

$$m_{\text{z'}}c_p\frac{\mathrm{d}T_{\text{z'}}}{\mathrm{d}t} = \dot{q}_{\text{gen,z'}} - \frac{k_{\text{z'x}}A_{\text{z'x}}}{L_{\text{z'x}}}(T_{\text{z'}} - T_{\text{x}}) - \frac{k_{\text{z'z}}A_{\text{z'z}}}{L_{\text{z'z}}}(T_{\text{z'}} - T_{\text{z}}) \tag{3}$$

$$m_{\text{z}}c_p\frac{\mathrm{d}T_{\text{z}}}{\mathrm{d}t} = \frac{k_{\text{z'z}}A_{\text{z'z}}}{L_{\text{z'z}}}(T_{\text{z'}} - T_{\text{z}}) - \frac{k_{\text{zx}}A_{\text{zx}}}{L_{\text{zx}}}(T_{\text{z}} - T_{\text{x}}) - h_{\text{z}\infty}A_{\text{z}\infty}(T_{\text{z}} - T_{\infty}) \tag{4}$$

The number of variables in the above equations can be significantly reduced by defining six heat diffusion time constants of the form given in Eq. (5) and (6), as well as two heat dissipation time constants as given in Eq. (7) and (8).

$$\tau_{\text{z'x}} = \frac{m_{\text{z'}}c_pL_{\text{z'x}}}{k_{\text{z'x}}A_{\text{z'x}}} \tag{5}$$

$$\tau_{\text{z'z}} = \frac{m_{\text{z'}}c_pL_{\text{z'z}}}{k_{\text{z'z}}A_{\text{z'z}}} \tag{6}$$

$$\tau_{\text{x}\infty} = \frac{m_{\text{x}}c_p}{h_{\text{x}}A_{\text{x}\infty}} \tag{7}$$

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$$\tau_{z\infty} = \frac{m_z c_p}{h_z A_{z\infty}} \tag{8}$$

The proposed model in Figure 4 is intentionally drawn such that the following physical property equivalencies can be applied: $L_{xz'} = L_{z'x}$, $L_{xz} = L_{zx}$, $L_{zz'} = L_{z'z}$, $A_{xz'} = A_{z'x}$, $A_{xz} = A_{zx}$, $A_{zz'} = A_{z'z}$, $k_{xz'} = k_{z'x}$, $k_{xz} = k_{zx}$, $k_{zz'} = k_{z'z}$, and $\frac{A_{zx}}{A_{z'x}} = \frac{m_z}{m_{z'}}$. These equivalencies enable two heat diffusion time constant equivalencies as shown in Eq. (9) and (10).

$$\tau_{xz'} = \tau_{xz} = \frac{m_x}{m_{z'}}\tau_{z'x} = \frac{m_x}{m_z}\tau_{zx} = \frac{m_x}{m_{z'}}\frac{m_{z'}}{m_z}\tau_{zx} \tag{9}$$

$$\tau_{z'z} = \frac{m_{z'}}{m_z}\tau_{zz'} \tag{10}$$

The thermal model therefore only requires two heat diffusion time constants $\tau_{z'x}$ and $\tau_{z'z}$ and two mass ratios $\frac{m_x}{m_{z'}}$ and $\frac{m_{z'}}{m_z}$, instead of six heat diffusion time constants. The two heat dissipation time constants $\tau_{x\infty}$ and $\tau_{z\infty}$ are also required. This gives a total of six parameters to describe the system in Figure 4. In the remainder of this article, the parameters λ_x and λ_z will be used to denote the two mass ratios, as defined by Eq. (11) and (12).

$$\lambda_x = \frac{m_x}{m_{z'}} \tag{11}$$

$$\lambda_z = \frac{m_{z'}}{m_z} \tag{12}$$

Additionally, the variables Θ_x and $\Theta_{z'}$ will be used to denote the cell heat generation rates of cell X and cell Z per unit heat capacity, as defined by Eq. (13) and (14), respectively. These quantities will hereafter be referred to as the 'adiabatic temperature rates' since they represent the temperature rate of change that would result from cell heat generation in a hypothetical adiabatic condition.

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$$\Theta_{\mathrm{x}} = \frac{\dot{q}_{\mathrm{gen,x}}}{m_{\mathrm{x}} c_p} \tag{13}$$

$$\Theta_{\mathrm{z}'} = \frac{\dot{q}_{\mathrm{gen,z}'}}{m_{\mathrm{z}'} c_p} \tag{14}$$

Using the definitions above in Eq. (5) through (14), the three system equations can be rewritten as Eq. (15) through (17).

$$\frac{\mathrm{d}T_{\mathrm{x}}}{\mathrm{d}t} = \Theta_{\mathrm{x}} + \frac{T_{\mathrm{z}} - T_{\mathrm{x}}}{\lambda_{\mathrm{x}} \lambda_{\mathrm{z}} \tau_{\mathrm{z}'\mathrm{x}}} + \frac{T_{\mathrm{z}'} - T_{\mathrm{x}}}{\lambda_{\mathrm{x}} \tau_{\mathrm{z}'\mathrm{x}}} - \frac{T_{\mathrm{x}} - T_{\infty}}{\tau_{\mathrm{x}\infty}} \tag{15}$$

$$\frac{\mathrm{d}T_{\mathrm{z}'}}{\mathrm{d}t} = \Theta_{\mathrm{z}'} - \frac{T_{\mathrm{z}'} - T_{\mathrm{x}}}{\tau_{\mathrm{z}'\mathrm{x}}} - \frac{T_{\mathrm{z}'} - T_{\mathrm{z}}}{\tau_{\mathrm{z}'\mathrm{z}}} \tag{16}$$

$$\frac{\mathrm{d}T_{\mathrm{z}}}{\mathrm{d}t} = \lambda_{\mathrm{z}} \frac{T_{\mathrm{z}'} - T_{\mathrm{x}}}{\tau_{\mathrm{z}'\mathrm{z}}} - \frac{T_{\mathrm{z}} - T_{\mathrm{x}}}{\tau_{\mathrm{z}'\mathrm{x}}} - \frac{T_{\mathrm{z}} - T_{\infty}}{\tau_{\mathrm{z}\infty}} \tag{17}$$

2.4. Numerical solutions

To achieve the simplest solutions to the three system equations, it is preferable to express the equations in the simplified forms shown in Eq. (18) through (21).

$$\frac{\mathrm{d}T_{\mathrm{x}}}{\mathrm{d}t} = \Theta_{\mathrm{x}} - \frac{T_{\mathrm{x}} - T_{\infty}}{\tau_{\mathrm{x}}} \tag{18}$$

$$\frac{\mathrm{d}T_{\mathrm{z}'}}{\mathrm{d}t} = \Theta_{z'} - \frac{T_{\mathrm{z}'} - T_{\infty}}{\tau_{\mathrm{z}'}} \tag{19}$$

$$\frac{\mathrm{d}T_{\mathrm{z}}}{\mathrm{d}t} = - \frac{T_{\mathrm{z}} - T_{\infty}}{\tau_{\mathrm{z}}} \tag{20}$$

These simplified equations require τ_{x}, $\tau_{\mathrm{z}'}$, and τ_z to have the definitions given in Eq. (21), Eq. (22), and Eq. (23), respectively. These quantities will be referred to as 'time variables' to distinguish them from 'time constants'. The time variables are a function of temperature as well as the time constants and mass ratios defined in Section 2.3.

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$$\frac{1}{\tau_{\mathrm{x}}}=\frac{1}{\tau_{\mathrm{x}\infty}}-\frac{1}{\lambda_{\mathrm{x}}\tau_{\mathrm{z}'\mathrm{x}}}\left(\frac{T_{\mathrm{z}'}-T_{\mathrm{x}}}{T_{\mathrm{x}}-T_{\infty}}\right)-\frac{1}{\lambda_{\mathrm{x}}\lambda_{\mathrm{z}}\tau_{\mathrm{z}'\mathrm{x}}}\left(\frac{T_{\mathrm{z}}-T_{\mathrm{x}}}{T_{\mathrm{x}}-T_{\infty}}\right) \tag{21}$$

$$\frac{1}{\tau_{\mathrm{z}'}}=\frac{1}{\tau_{\mathrm{z}'\mathrm{x}}}\left(\frac{T_{\mathrm{z}'}-T_{\mathrm{x}}}{T_{\mathrm{z}'}-T_{\infty}}\right)+\frac{1}{\tau_{\mathrm{z}'\mathrm{z}}}\left(\frac{T_{\mathrm{z}'}-T_{\mathrm{z}}}{T_{\mathrm{z}'}-T_{\infty}}\right) \tag{22}$$

$$\frac{1}{\tau_{\mathrm{z}}}=\frac{1}{\tau_{\mathrm{z}\infty}}+\frac{1}{\tau_{\mathrm{z}'\mathrm{x}}}\left(\frac{T_{\mathrm{z}}-T_{\mathrm{x}}}{T_{\mathrm{z}}-T_{\infty}}\right)-\frac{\lambda_{\mathrm{z}}}{\tau_{\mathrm{z}'\mathrm{z}}}\left(\frac{T_{\mathrm{z}'}-T_{\mathrm{z}}}{T_{\mathrm{z}}-T_{\infty}}\right) \tag{23}$$

Analytical solutions can be derived for the three simplified system equations in Eq. (18) through (21). The analytical steps are detailed in Section S1 of the Supplementary Materials. The final results are shown in Eq. (24) through (26) below, where T_{∞} is assumed to non-constant and the zero subscript indicates an initial condition at time $t=0$.

$$T_{\mathrm{x}}=\mathrm{e}^{-\frac{t}{\tau_{\mathrm{x}}}}\left(T_{\mathrm{x},0}+\int_0^t \mathrm{e}^{\frac{t}{\tau_{\mathrm{x}}}}\left(\frac{1}{\tau_{\mathrm{x}}}T_{\infty}+\Theta_{\mathrm{x}}\right)\mathrm{d}t\right) \tag{24}$$

$$T_{\mathrm{z}'}=\mathrm{e}^{-\frac{t}{\tau_{\mathrm{z}'}}}\left(T_{\mathrm{z}',0}+\int_0^t \mathrm{e}^{\frac{t}{\tau_{\mathrm{z}'}}}\left(\frac{1}{\tau_{\mathrm{z}'}}T_{\infty}+\Theta_{\mathrm{z}'}\right)\mathrm{d}t\right) \tag{25}$$

$$T_{\mathrm{z}}=\mathrm{e}^{-\frac{t}{\tau_{\mathrm{z}}}}\left(T_{\mathrm{z},0}+\int_0^t \frac{1}{\tau_{\mathrm{z}}}T_{\infty}\mathrm{e}^{\frac{t}{\tau_{\mathrm{z}}}}\mathrm{d}t\right) \tag{26}$$

Note that these solutions can only be derived analytically if τ_{x}, τ_{x}, and τ_{x} are treated as constants. Since these 'time variables' are actually non-constant, Eq. (24) through (26) are not true analytical solutions. However, these solutions provide a starting point from which functional numerical solutions can be developed. The next step is to convert the solutions from continuous form to discretized form, as shown in Eq. (27) through (29).

$$T_{\mathrm{x},j}=\mathrm{e}^{-\frac{t_j}{\tau_{\mathrm{x},j}}}\left(T_{\mathrm{x},0}+\sum_{i=1}^{j}\mathrm{e}^{\frac{t_i}{\tau_{\mathrm{x},i}}}\left(\frac{1}{\tau_{\mathrm{x},i}}T_{\infty,i}+\Theta_{\mathrm{x},i}\right)\Delta t_i\right) \tag{27}$$

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$$T_{z',j} = e^{-\frac{t_j}{\tau_{z',j}}}\left(T_{z',0} + \sum_{i=1}^{j} e^{\frac{t_i}{\tau_{z',i}}}\left(\frac{1}{\tau_{z',i}}T_{\infty,i} + \Theta_{z',i}\right)\Delta t_i\right) \quad (28)$$

$$T_{x,j} = e^{-\frac{t_j}{\tau_{z,j}}}\left(T_{z,0} + \sum_{i=1}^{j} e^{\frac{t_i}{\tau_{z,i}}}\left(\frac{1}{\tau_{z,i}}T_{\infty,i}\right)\Delta t_i\right) \quad (29)$$

These discretized equations require a summation of calculated values from time interval $i = 1$ to time interval $i = j$ in order to calculate the temperature for interval j. This method is computationally expensive to repeat over an indefinite series of real-time datapoints, as would be the case in a commercial second-life battery project. To reduce the computational requirements, the discretized equations can be converted to a functional recursive form. The derived recursive system equations are shown in Eq. (30) through (32). The complete conversion steps are detailed in Section S2 of the Supplementary Materials.

$$T_{x,i} = T_{x,i-1}e^{-\frac{\Delta t_i}{\tau_{x,i}}} + \frac{\Delta t_i}{\tau_{x,i}}T_{\infty,i} + \Theta_{x,i}\Delta t_i \quad (30)$$

$$T_{z',i} = T_{z',i-1}e^{-\frac{\Delta t_i}{\tau_{z',i}}} + \frac{\Delta t_i}{\tau_{z',i}}T_{\infty,i} + \Theta_{z',i}\Delta t_i \quad (31)$$

$$T_{z,i} = T_{z,i-1}e^{-\frac{\Delta t_i}{\tau_{z,i}}} + \frac{\Delta t_i}{\tau_{z,i}}T_{\infty,i} \quad (32)$$

2.5. Heat generation approximation

The proposed VTS model seeks to describe T_x as a function of T_z and T_∞ measurements, but Eq. (30) through (32) show that T_x also depends on Θ_x and $\Theta_{z'}$ (since τ_x is a function of $T_{z'}$). The VTS model therefore requires that Θ_x and $\Theta_{z'}$ be describable in terms of the measured T_z and T_∞. Such a definition for $\Theta_{z'}$ can be approximated by rearranging Eq. (19) and converting to discrete form using the finite difference method. These steps are carried out in Eq. (33) and (34).

$$\Theta_{z'} = \frac{dT_{z'}}{dt} + \frac{T_{z'} - T_\infty}{\tau_{z'}} \quad (33)$$

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$$\Theta_{z',i} \approx \frac{T_{z',i} - T_{z',i-1}}{\Delta t_i} + \frac{T_{z',i} - T_{\infty,i}}{\tau_{z',i}} \tag{34}$$

Eq. (34) is not useful since $T_{z'}$ is unknown. However, the measured value T_z^{meas} can be used to apply the numerical approximation shown in Eq. (35). Note that T_∞ will henceforth be written as T_∞^{meas} in the equations so that all measured temperatures are denoted as such. The variable α_Θ is a correction factor which is described in Section 2.7.

$$\Theta_{z',i} \approx \alpha_\Theta \left(\frac{T_{z,i}^{\text{meas}} - T_{z,i-1}^{\text{meas}}}{\Delta t_i} + \frac{T_{z,i}^{\text{meas}} - T_{\infty,i}^{\text{meas}}}{\tau_{z',i}} \right) \tag{35}$$

The validity of Eq. (35) relies on the assumption that T_z and $T_{z'}$ exhibit similar rates of change. This is a reasonable assumption since T_z and $T_{z'}$ belong to the same cell and are only separated by axial heat diffusion, which is relatively fast in cylindrical lithium-ion cells [14]. However, it should be recognized that the validity of the assumption may deteriorate at very high heat generation rates or large step changes in heat generation (e.g. beginning or end of discharge or charge).

With $\Theta_{z'}$ now adequately described in terms of known values, the VTS model requires an additional equation to relate Θ_x and $\Theta_{z'}$. Based on the equation for the total reversible and irreversible heat generation rate of a cell [9], Θ can be redefined as Eq. (36), where I is the cell current ($I < 0$ for discharge), V is the cell voltage, V_{rev} is the cell reversible voltage, and $\frac{\partial V_{\text{rev}}}{\partial T}$ is the cell entropic heat coefficient.

$$\Theta = \frac{\dot{q}_{\text{gen}}}{mc_p} = \frac{I(V - V_{\text{rev}}) + IT\dfrac{\partial V_{\text{rev}}}{\partial T}}{mc_p} \tag{36}$$

The relationship between Θ_x and $\Theta_{z'}$ is therefore given by Eq. (37).

$$\frac{\Theta_x}{\Theta_{z'}} = \frac{m_{z'}\dot{q}_{\text{gen},x}}{m_x\dot{q}_{\text{gen},z'}} = \frac{m_{z'}\left(I_x(V_x - V_{\text{rev}}) + I_x T_x \dfrac{\partial V_{\text{rev}}}{\partial T}\right)}{m_x\left(I_{z'}(V_{z'} - V_{\text{rev}}) + I_{z'} T_{z'} \dfrac{\partial V_{\text{rev}}}{\partial T}\right)} \tag{37}$$

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The quantity $V - V_{\text{rev}}$ depends on the cell internal resistance, which varies with temperature, SOC, and SOH. For aged second-life batteries, it is especially likely that any two series-connected cell groups in the battery pack will exhibit different voltages, but the BMS provides this voltage information. Current is equivalent in series-connected cells, but since each cell group in the Tesla Model 3 SR+ battery pack consists of 31 parallel-connected cells, it is not necessarily true that $I_{\text{x}} = I_{\text{z}'}$, and the BMS does not provide this current distribution information. The above equation also requires that V_{rev} and $\frac{\partial V_{\text{rev}}}{\partial T}$ are known as a function of SOC, and this information is rarely available for a given EV battery pack unless lengthy testing is carried out over a range of controlled temperatures in a specialized test chamber.

For this pilot study, it is assumed that $\dot{q}_{\text{gen,x}} \approx \dot{q}_{\text{gen,z}'}$, which reduces Eq. (37) to $\frac{\Theta_{\text{x},i}}{\Theta_{\text{z}',i}} \approx \frac{m_{\text{z}'}}{m_{\text{x}}} \approx \frac{1}{\lambda_{\text{x}}}$. Based on this assumption, Θ_{x} can be approximated using $T_{\text{z}}^{\text{meas}}$ according to Eq. (38). This assumption will prove adequate for this particular VTS model, but it may not always be an appropriate assumption. This will be discussed further in Section 5.1.

$$\Theta_{\text{x},i} \approx \frac{\alpha_\Theta}{\lambda_{\text{x}}}\left(\frac{T_{\text{z},i}^{\text{meas}} - T_{\text{z},i-1}^{\text{meas}}}{\Delta t_i} + \frac{T_{\text{z},i}^{\text{meas}} - T_{\infty,i}^{\text{meas}}}{\tau_{\text{z}',i}}\right) \tag{38}$$

The definitions of Θ_{x} and $\Theta_{\text{z}'}$ are applied to Eq. (30) through (32) to provide a final set of recursive system equations for the three temperatures in the VTS model. The final equations are given in Eq. (39) through (41).

$$T_{\text{x},i} = T_{\text{x},i-1}\text{e}^{-\frac{\Delta t_i}{\tau_{\text{x},i}}} + \frac{\Delta t_i}{\tau_{\text{x},i}} T_{\infty,i}^{\text{meas}} + \frac{\alpha_\Theta}{\lambda_{\text{x}}}\left(T_{\text{z},i}^{\text{meas}} - T_{\text{z},i-1}^{\text{meas}} + \frac{\Delta t_i}{\tau_{\text{z}',i}}\left(T_{\text{z},i}^{\text{meas}} - T_{\infty,i}^{\text{meas}}\right)\right) \tag{39}$$

$$T_{\text{z}',i} = T_{\text{z}',i-1}\text{e}^{-\frac{\Delta t_i}{\tau_{\text{z}',i}}} + \frac{\Delta t_i}{\tau_{\text{z}',i}} T_{\infty,i}^{\text{meas}} + \alpha_\Theta\left(T_{\text{z},i}^{\text{meas}} - T_{\text{z},i-1}^{\text{meas}} + \frac{\Delta t_i}{\tau_{\text{z}',i}}\left(T_{\text{z},i}^{\text{meas}} - T_{\infty,i}^{\text{meas}}\right)\right) \tag{40}$$

$$T_{\text{z},i} = T_{\text{z},i-1}\text{e}^{-\frac{\Delta t_i}{\tau_{\text{z},i}}} + \frac{\Delta t_i}{\tau_{\text{z},i}} T_{\infty,i}^{\text{meas}} \tag{41}$$

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2.6. Heat dissipation parameters

The system equations presented in Section 2.3 include four time constants: $\tau_{z'x}$ to model lateral heat diffusion from cell Z to cell X, $\tau_{z'z}$ to model axial heat diffusion within cell Z, $\tau_{z\infty}$ to model heat dissipation from cell Z, and $\tau_{x\infty}$ to model heat dissipation from cell X. For heat diffusion (i.e. conduction), $\tau_{z'x}$ and $\tau_{z'z}$ depend on geometry, heat capacity, and thermal conductivity. Though thermal conductivity and specific heat capacity vary with temperature, the dependence is weak enough [14] that both can be treated as constant values over a typical operational battery temperature range without detriment to the VTS model. This means that $\tau_{z'x}$ and $\tau_{z'z}$ can be modelled as constant parameters. For heat dissipation (i.e. convection and radiation), $\tau_{z\infty}$ and $\tau_{x\infty}$ depend on geometry, heat capacity, emissivity $\varepsilon_{\mathrm{rad}}$, and convection coefficient h_{conv}. It is well known that h_{conv} can have a strong dependence on temperature in the case of natural convection [15]. Since this VTS model deals with a battery cooled passively by natural convection, the corresponding temperature dependence of $\tau_{z\infty}$ and $\tau_{x\infty}$ must be accounted for in the model.

Eq. (42) and (43) have been derived to give temperature-dependent definitions of $\tau_{z\infty}$ and $\tau_{x\infty}$ for this VTS model. The detailed derivations are provided in Section S3 of the Supplementary Materials, accounting for pack geometry, Rayleigh number, Prandtl number, Nusselt number, and the combined effects of convection and radiation. It is important to note that the values of $\tau_{z\infty}$ and $\tau_{x\infty}$ for a given time interval i are calculated using constant reference values $\tau^*_{z\infty}$ and $\tau^*_{x\infty}$, constant parameter $\varepsilon_{\mathrm{rad}}$, and the temperature values of the previous time interval $i-1$.

$$\tau_{z\infty,i} = \frac{\tau^*_{z\infty}}{(0.43 + 0.31\varepsilon_{\mathrm{rad}})\left(T_{z,i-1} - T^{\mathrm{meas}}_{\infty,i-1}\right)^{(0.66-0.20\varepsilon_{\mathrm{rad}})^{2.81}}} \tag{42}$$

$$\tau_{x\infty,i} = \frac{\tau^*_{x\infty}}{(0.54 + 0.25\varepsilon_{\mathrm{rad}})\left(T_{x,i-1} - T^{\mathrm{meas}}_{\infty,i-1}\right)^{(0.50-0.18\varepsilon_{\mathrm{rad}})^{2.14}}} \tag{43}$$

With these expressions of $\tau_{z\infty}$ and $\tau_{x\infty}$, the VTS model now features $\tau^*_{z\infty}$ and $\tau^*_{x\infty}$ as constant system parameters. These expressions also introduce $\varepsilon_{\mathrm{rad}}$ as an additional system parameter which can be used to explore the significance of radiation vs convection. The complete list of seven system

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parameters is $\tau_{z'x}$, $\tau_{z'z}$, $\tau^*_{z\infty}$, $\tau^*_{x\infty}$, λ_z, λ_x, and ε_{rad}, all of which are constants to be determined for this particular VTS model.

2.7. Functionality

A few final steps are required to make the three system equations functional for the VTS model. One unresolved issue is that the time variables τ_x, τ_z, and $\tau_{z'}$ must be calculated from T_x, T_z, and $T_{z'}$. This issue can be overcome by employing the finite difference method, whereby the values of τ_x, τ_z, and $\tau_{z'}$ in time interval i are calculated using the temperatures in the previous time interval $i-1$. This method is expressed in Eq. (44) through (46). Note that these equations use the modelled T_z instead of the measured T_z^{meas} in order to maintain energy conservation within the model.

$$\tau_{x,i} = \frac{1}{\frac{1}{\tau_{x\infty,i}} - \frac{1}{\lambda_x \tau_{z'x}}\left(\frac{T_{z',i-1} - T_{x,i-1}}{T_{x,i-1} - T^{meas}_{\infty,i-1}}\right) - \frac{1}{\lambda_x \lambda_z \tau_{z'x}}\left(\frac{T_{z,i-1} - T_{x,i-1}}{T_{x,i-1} - T^{meas}_{\infty,i-1}}\right)} \tag{44}$$

$$\tau_{z',i} = \frac{1}{\frac{1}{\tau_{z'x}}\left(\frac{T_{z',i-1} - T_{x,i-1}}{T_{z',i-1} - T^{meas}_{\infty,i-1}}\right) + \frac{1}{\tau_{z'z}}\left(\frac{T_{z',i-1} - T_{z,i-1}}{T_{z',i-1} - T^{meas}_{\infty,i-1}}\right)} \tag{45}$$

$$\tau_{z,i} = \frac{1}{\frac{1}{\tau_{z\infty,i}} + \frac{1}{\tau_{z'x}}\left(\frac{T_{z,i-1} - T_{x,i-1}}{T_{z,i-1} - T^{meas}_{\infty,i-1}}\right) - \frac{\lambda_z}{\tau_{z'z}}\left(\frac{T_{z',i-1} - T_{z,i-1}}{T_{z,i-1} - T^{meas}_{\infty,i-1}}\right)} \tag{46}$$

The correction factor α_Θ in Eq. (35) is also required to make the model functional. This variable must be defined in a way that compensates for the errors that arise in the model due to $\Theta_{z'}$ being approximated from T_z^{meas}. When the battery is in standby mode ($I = 0$) there is no heat generation ($\Theta = 0$), so this condition requires $\alpha_\Theta = 0$. When the battery is operating ($|I| > 0$) both cell X and cell Z will generate heat ($\Theta > 0$), so this condition requires $\alpha_\Theta \geq 1$. The simplest scheme for α_Θ is to compare the calculated value $T_{z,i}$ to the measured value $T^{meas}_{z,i}$ for each time interval i, and use this comparison to determine the value of $\alpha_{\Theta,i}$ for that time interval. While proportional, integral, or derivative-based controls could be applied to adjust α_Θ during battery operation, for this VTS model it is sufficient to use a fixed $\alpha_\Theta \geq 1$ value whenever $T_{z,i} < T^{meas}_{z,i}$, and otherwise set $\alpha_{\Theta,i} =$

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1. This allows for a simple VTS model which requires no iteration whatsoever. By trial and error, the accuracy of the VTS model was found to be fairly insensitive to α_θ values between 1.1 and 1.5, with $\alpha_\theta = 1.2$ giving optimal results. The full definition of α_θ is given in Eq. (47).

$$\alpha_{\theta,i} = \begin{cases} 0 \ , & I_i = 0 \\ 1 \ , & |I_i| > 0,\ T_{\mathrm{z},i} \geq T_{\mathrm{z},i}^{\mathrm{meas}} \\ 1.2 \ , & |I_i| > 0,\ T_{\mathrm{z},i} < T_{\mathrm{z},i}^{\mathrm{meas}} \end{cases} \tag{47}$$

The overall functionality of this VTS model is visualized in Figure 5, showing how the present value of T_{x} is determined using measured data, a set of seven constant parameters, and the calculated results of the previous time interval.

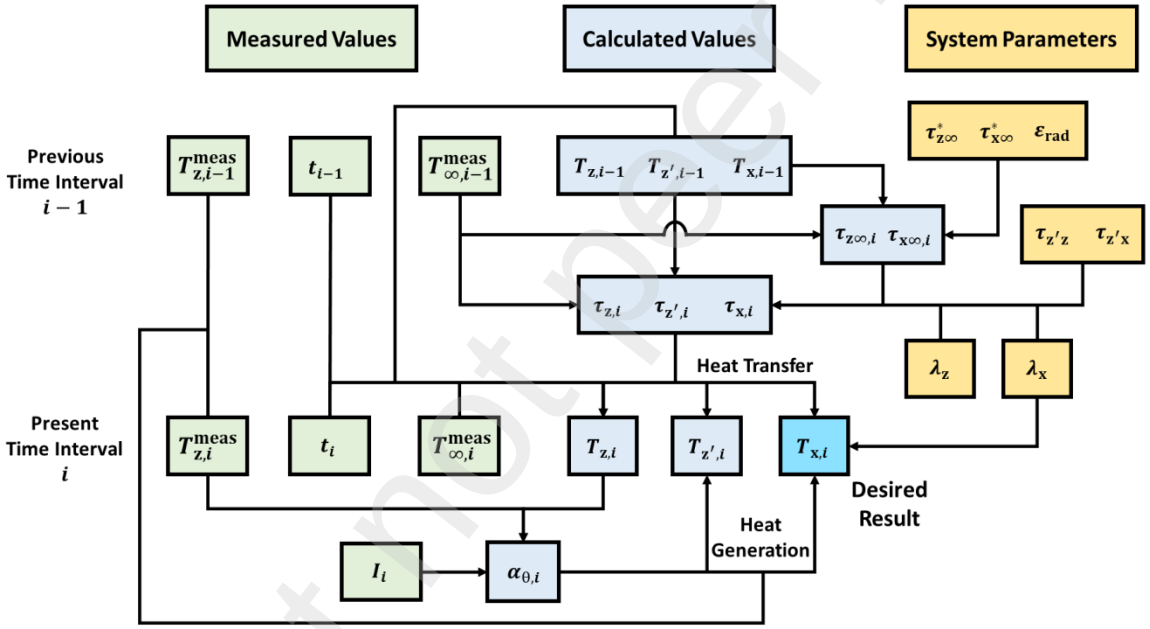


Figure 5 Visualization of the proposed VTS model for a Tesla Model 3 SR+ battery pack.

The final requirement of the VTS model is a set of initial conditions for the first interval at $i = 0$, specifically $T_{\mathrm{z},0}$, $T_{\mathrm{z}',0}$, and $T_{\mathrm{x},0}$. While it is straightforward to equate $T_{\mathrm{z},0}$ to its measured value $T_{\mathrm{z},0}^{\mathrm{meas}}$, initial guesses must be made for the unknown $T_{\mathrm{z}',0}$ and $T_{\mathrm{x},0}$. If the battery has been idle in a consistent ambient environment for a sufficient period of time (e.g. one day or more) then the battery temperature should be close to a uniformity condition at $T_{\infty,0}^{\mathrm{meas}}$. However, a more general definition of initial conditions should be established to account for cases where the uniformity condition is not met. In this particular VTS model, Figure 4 implies $T_{\mathrm{z}'} \geq T_{\mathrm{z}} \geq T_{\mathrm{x}} \geq T_{\infty}$, and it is

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expected that $T_{z'}$ should remain fairly close to T_z under most circumstances. Eq. (48) through (50) therefore provide a reasonable set of generalized initial conditions for this VTS model.

$$T_{z,0} = T_{z,0}^{meas} \tag{48}$$

$$T_{z',0} = T_{z,0}^{meas} \tag{49}$$

$$T_{x,0} = \frac{T_{z,0}^{meas} + T_{\infty,0}^{meas}}{2} \tag{50}$$

Even if there is a significant error in these initial guesses, the exponential terms in Eq. (39) through (41) will cause the error to dissipate from the model in a relatively short amount of time (e.g. a few hours). This practical feature of the VTS model will be discussed further in Section 4.2.

Altogether, the computational methods proposed in this section can be executed in a standard Microsoft Excel spreadsheet software without any need for iterative modeling software. The simplicity of the execution is an attractive feature of the VTS model.

3. Parameter optimization

This VTS model is designed with seven constant parameters which can be used to describe an unknown T_x as a function of measured T_z^{meas} and T_∞^{meas} using Eq. (39) through (50). The numerical values of these constant parameters must be determined empirically using experimental training data featuring measurements of T_z^{meas} and T_∞^{meas} as well as the desired temperature T_x^{meas}. Numerical parameter optimization techniques can be employed to determine a set of parameter values which optimizes the fit between the calculated temperatures of the VTS model and the measured temperatures of the empirical dataset. This section will describe the methods used to carry out parameter optimization for this particular VTS model.

3.1. Empirical training data

Acquiring an experimental dataset for the unknown T_x^{meas} necessitates partial pack disassembly and installation of a temperature sensor on cell X. If a VTS model can be established for a single EV battery pack of a given design, the model should be applicable to any battery pack of the same design without the need for pack disassembly. The VTS concept therefore requires partial

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disassembly of a single 'pilot pack' to enable enhanced thermal monitoring for an unlimited number of identical packs. One non-invasive alternative would be a thermal camera, which uses infrared imaging to detect the temperature of an object. However, this method can only measure the external surface temperature of the battery pack housing. Direct measurement with retrofitted temperature sensors at desired locations inside the battery pack should yield the most accurate results. Figure 6 shows the placement of a T-type thermocouple on the exposed cylindrical wall surface of cell X at the corner of an outer module after removal of the pack housing cover. For measurement consistency, an identical T-type thermocouple is also placed on the exposed upper surface of cell Z (see Figure 2) to acquire $T_{\mathrm{z}}^{\mathrm{meas}}$, instead of using Tesla's built-in thermistors. A third identical T-type thermocouple is placed in open air about 3 m away from the pack to acquire $T_{\infty}^{\mathrm{meas}}$.

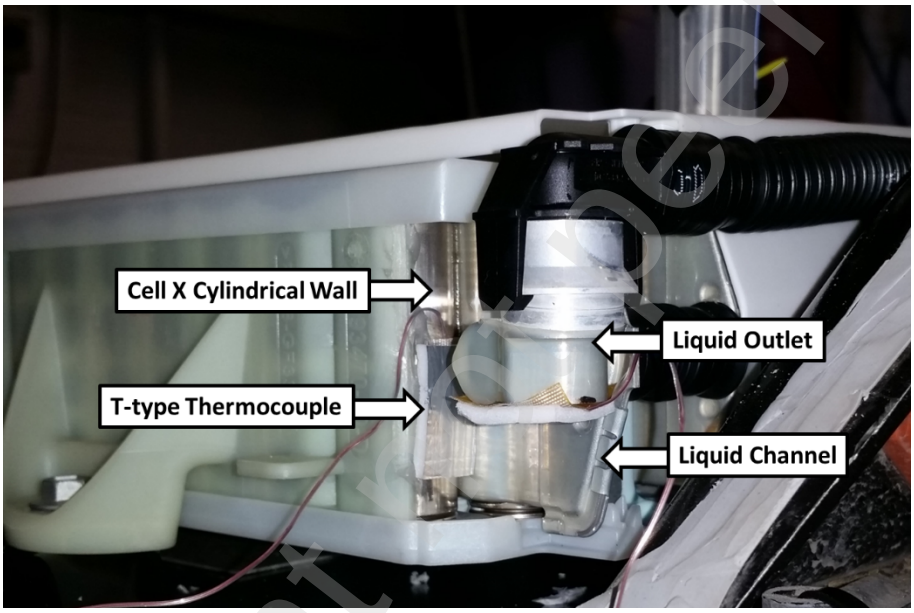


Figure 6 Cell X temperature sensor placement for acquisition of empirical training data.

Since second-life batteries will operate under a range of power signals and ambient conditions, a VTS model must be robust to ensure accuracy is maintained in any operational scenario. The empirical data used to determine the system parameters should therefore encompass the complete range of operational scenarios that are expected in a second-life energy storage application. This would entail testing the battery at low, medium, and high power, at constant and variable power, and in cold, temperate, and warm test environments. For this pilot study, a 2019 Tesla Model 3 SR+ battery pack was subjected to deep-discharge tests at constant-power with lower and upper

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cell voltage limits of 2.5 V and 4.2 V, respectively. Three tests were conducted to capture thermal performance at low, medium, and high power rates in a relatively consistent ambient temperature near $T_\infty =$ 19 °C. The three measured datasets are used for empirical 'training' of the VTS model to determine optimal values for the seven constant parameters. While the robustness of the VTS model in diverse ambient conditions cannot be explored using these particular datasets, robustness in different power ranges can be demonstrated.

The empirical datasets are shown in Figure 7. There is clearly a significant difference between T_x and T_z at all three power rates, with a more pronounced difference at higher power rates. Without a VTS model, this significant difference would go undetected if a Tesla Model 3 battery pack were deployed as-received in a passive-cooled second-life energy storage application, which could lead to thermal mismanagement resulting in degradation divergence and decreased pack lifetime.

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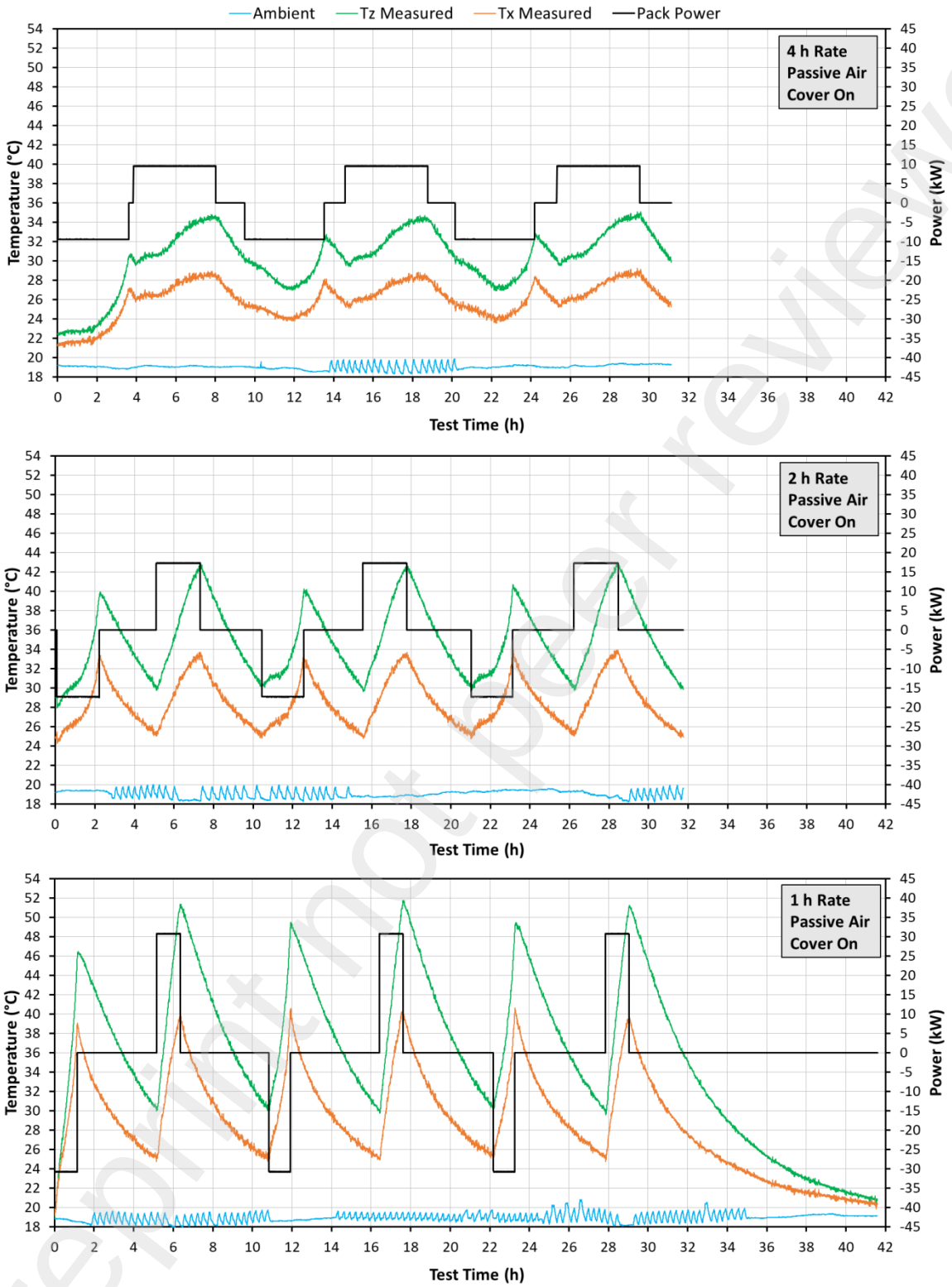


Figure 7 Experimental dataset for empirical determination of VTS model parameters, including 4 h (top), 2 h (middle), and 1 h (bottom) constant power tests conducted on a 2019 Tesla Model 3 SR+ battery pack.

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Note that since the system equations employ the finite difference method, the effectiveness of the model depends on the data sampling rate. The empirical dataset in Figure 7 was originally measured using a 1 s sampling interval, but the raw data was post-processed to adjust the data time interval Δt for application in the VTS model. It was found that the VTS model works best when 100 or more datapoints are available for a complete discharge or charge. This corresponds to maximum data intervals of 36 s, 72 s, and 144 s for a 1 h rate, 2 h rate, and 4 h rate, respectively. For this VTS model, $\Delta t =$ 10 s was selected since this gives high-quality results without unnecessary computational requirements.

3.2. Curve fitting

A variety of parameter optimization algorithms are available in different software languages. For this VTS model, parameter optimization was carried out using the generalized reduced gradient nonlinear solving method within Microsoft Excel Solver, developed by Frontline Systems Inc. [16]. This is a free optimization tool which operates within a standard Microsoft Excel spreadsheet. The software requires specification of a single calculated 'objective value' to be either minimized or maximized, as well as one or more 'decision variables' which the algorithm adjusts (within defined constraints) in order to change the objective value. The algorithm determines the optimal set of decision variables that minimizes or maximizes the objective value. For this VTS model, the seven constant parameters are specified as the decision variables, and the error of the VTS model is specified as the objective value to be minimized.

Since the VTS model calculates values for both T_{z} and T_{x}, both these values can be compared to their measured values $T_{\mathrm{z}}^{\mathrm{meas}}$ and $T_{\mathrm{x}}^{\mathrm{meas}}$ to calculate the total model error. The squared errors for T_{z} and T_{x} are calculated as the squared error in each time interval i averaged from the beginning ($i = 0$) to the end ($i = i_{\mathrm{end}}$) of the full dataset as shown in Eq. (51) and (52). The total error for the VTS model Err_{VTS} is then taken as the sum of the two individual errors Err_{z} and Err_{x} as shown in Eq. (53). Note that the full dataset refers to the combination of the three empirical datasets shown in Figure 7.

$$(Err_{\mathrm{z}})^2 = \frac{1}{i_{\mathrm{end}} + 1} \sum_{i=0}^{i_{\mathrm{end}}} \left(T_{\mathrm{z},i} - T_{\mathrm{z},i}^{\mathrm{meas}}\right)^2 \tag{51}$$

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$$(Err_{\mathrm{x}})^2 = \frac{1}{i_{\mathrm{end}}+1}\sum_{i=0}^{i_{\mathrm{end}}}(T_{\mathrm{x},i} - T_{\mathrm{x},i}^{\mathrm{meas}})^2 \quad (52)$$

$$Err_{\mathrm{VTS}} = Err_{\mathrm{z}} + Err_{\mathrm{x}} \quad (53)$$

For parameter optimization, the initial guess for T_{x} is set at $T_{\mathrm{x},0} = T_{\mathrm{x},0}^{\mathrm{meas}}$ so that the optimization is not influenced by any initial error. The impact of error in the initial guess on the VTS model results will be explored in Section 4.2.

3.3. Optimization constraints

Parameter optimization typically requires an initial estimate as well as constraints (e.g. lower and upper bounds) for each parameter in a given model. In theory, the true values of all seven parameters in this VTS model could be calculated based on the physical properties of the battery pack and surrounding environment. However, the complexity of the Tesla Model 3 SR+ battery pack makes it challenging to determine these values with confidence. Furthermore, since the thermal model underpinning the VTS model (illustrated in Figure 4) is highly simplified, the optimized values of the seven parameters will not necessarily reflect their true values. The seven parameters should be thought of as *effective* properties of the battery pack which enable the simplified thermal model to accurately reproduce empirical measurements. For example, the value of λ_{z} does necessarily represent the true ratio of the cell stack mass to the cell housing mass, but rather gives the *effective* ratio required to model axial heat diffusion within the cell using the simplified thermal model in Figure 4. Even so, a well-designed VTS model should give parameter values that are within a reasonable range of the expected true values. To facilitate comparison between the model parameters and physical expectations, each parameter is converted to a corresponding 'physical parameter' using mathematical relationships described in Section S4 of the Supplementary Materials.

Table 1 summarizes the estimated value, lower bound, and upper bound for each physical parameter, with the equivalent estimate for each VTS model parameter listed on the far right. Complete calculations and justifications for each value are provided in Section S4 of the Supplementary Materials. Note that $h_{\mathrm{conv,x}}^{*}$ and $h_{\mathrm{conv,z}}^{*}$ respectively represent reference values for the natural convection coefficients of cell X and cell Z. The other parameters in Table 1 have

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already been defined. The corresponding VTS model parameter estimates on the far right of Table 1 are used to initialize the decision variables in the Solver parameter optimization tool.

Table 1 Estimated values and ranges for physical parameters and VTS model parameters.

Parameter (physical)	Estimated Value	Lower Bound	Upper Bound	Parameter (VTS model)	Estimated Value
$m_{\text{cell}}c_p\left(\frac{\text{J}}{\text{K}}\right)$	67	59	78	$m_{\text{cell}}c_p\left(\frac{\text{J}}{\text{K}}\right)$	*result*
$\frac{m_{\text{z}'}}{m_{\text{cell}}}$	0.80	0.6	0.97	λ_{z}	4.0
$\frac{m_{\text{x}}}{m_{\text{cell}}}$	1.5	1.0	2.0	λ_{x}	1.9
ε_{rad}	0.50	0.0	1.0	ε_{rad}	0.50
$h^*_{\text{conv,z}}\left(\frac{\text{W}}{\text{m}^2\text{K}}\right)$	4.49	0.1	6	$\tau^*_{\text{z}\infty}$ (h)	0.94
$h^*_{\text{conv,x}}\left(\frac{\text{W}}{\text{m}^2\text{K}}\right)$	5.06	0.1	6	$\tau^*_{\text{x}\infty}$ (h)	0.87
$k_{\text{z}'\text{z}}\left(\frac{\text{W}}{\text{m}\cdot\text{K}}\right)$	30	10	50	$\tau_{\text{z}'\text{z}}$ (h)	0.050
$k_{\text{z}'\text{x}}\left(\frac{\text{W}}{\text{m}\cdot\text{K}}\right)$	1.0	0.1	5	$\tau_{\text{z}'\text{x}}$ (h)	3.6

In addition to the constraints applied to the seven parameters, Eq. (54) and (55) provide a novel means of constraining the parameter optimization based on the heat $q^{\text{cyc}}_{\text{gen}}$ generated by each cell during a complete discharge-charge cycle between fixed SOC limits. The $\Delta T^{\text{dis}}_{\text{ad}}$ and $\Delta T^{\text{chg}}_{\text{ad}}$ terms represent the 'adiabatic temperature change' that would be experienced by each cell during discharge and charge in a theoretical adiabatic condition where no heat can be exchanged with the surroundings. Section S5 of the Supplementary Materials describes how these quantities are calculated within the VTS model.

$$m_{\text{x}}c_p = \frac{q^{\text{cyc}}_{\text{gen,x}}}{\Delta T^{\text{dis}}_{\text{ad,x}} + \Delta T^{\text{chg}}_{\text{ad,x}}} \tag{54}$$

$$m_{\text{z}'}c_p = \frac{q^{\text{cyc}}_{\text{gen,z}'}}{\Delta T^{\text{dis}}_{\text{ad,z}'} + \Delta T^{\text{chg}}_{\text{ad,z}'}} \tag{55}$$

Since the empirical training data includes complete discharge-charge cycles at three different power rates, Eq. (54) and (55) will produce three values each for $m_{\text{x}}c_p$ and $m_{\text{z}'}c_p$. The heat capacity of a given cell should be consistent for all power rates, so the relative standard deviation of the calculated mc_p values will be used as a VTS model verification metric.

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The magnitude of the calculated mc_p values can also be compared to physical expectations. To establish expectations, a single cell from a Tesla Model 3 battery pack was acquired and weighed at $m_{\text{cell}} =$ 70 g. This can be related to the VTS model using Eq. (56).

$$m_{\text{cell}} = m_{\text{z}'} + m_{\text{z}} = \left(1 + \frac{1}{\lambda_{\text{z}}}\right) m_{\text{z}'} = 0.070 \text{ kg} \tag{56}$$

The 2022 review by Steinhardt et al. [14] reported c_p values for five cylindrical NCA/graphite cells in the range of 830–1133 $\frac{\text{J}}{\text{kg} \cdot \text{K}}$. This gives Eq. (57) as a constraint for the VTS model.

$$58\frac{\text{J}}{\text{K}} < m_{\text{cell}} c_p < 79\frac{\text{J}}{\text{K}} \tag{57}$$

Figure 8 illustrates the complete parameter optimization scheme including the role of the physical parameters and the mc_p outputs as optimization constraints. During parameter optimization, the present values of the seven VTS model parameters are converted to their corresponding physical parameters. These physical values are used to constrain the parameter optimization within the lower and upper bounds in Table 1. For example, Solver can only use values of $\tau_{\text{z}'\text{z}}$ that produce values of $k_{\text{z}'\text{z}}$ between 10 $\frac{\text{W}}{\text{m} \cdot \text{K}}$ and 50 $\frac{\text{W}}{\text{m} \cdot \text{K}}$ based on the specified $L_{\text{z}'\text{z}}$ and $A_{\text{z}'\text{z}}$ and the present result for $m_{z'}c_p$, (see Eq. (6) in Section 2.3), where $m_{z'}c_p$ is determined from the calculated values of $\Delta T^{\text{dis}}_{\text{ad,z}'}$ and $\Delta T^{\text{chg}}_{\text{ad,z}'}$.

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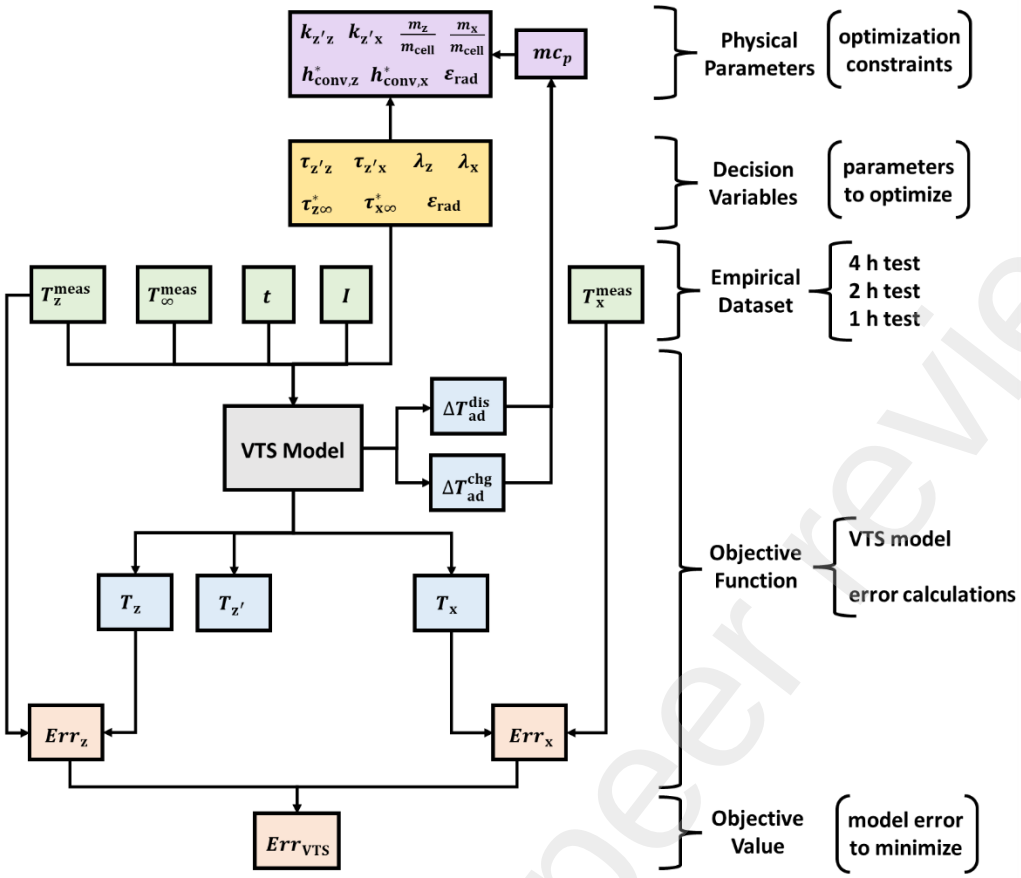


Figure 8 Parameter optimization scheme applied to the VTS model, including physical constraints.

4. Results

Parameter optimization revealed that there are many possible combinations of parameter values that can produce accurate results for this VTS model. Essentially, each parameter has a range within which the accuracy of the model can be retained by adjusting one or more of the other six parameters. This section will examine the effectiveness of the VTS model for the set of parameter values that was ultimately selected as the most appropriate. Section S6 of the Supplementary Materials include a detailed sensitivity analysis which explores the full range of parameter values that can be used without sacrificing VTS model accuracy.

4.1. Model accuracy

The best results achieved by the parameter optimization are shown in Figure 9. The modelled T_z and T_x are in very good agreement with the measured values throughout the entire test duration at

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all three power rates. This demonstrates that the simplified thermal model in Figure 4 is capable of accurately reproducing the empirical data when the right parameter values are assigned.

The T_{x} model shown in Figure 9 is highly accurate in nearly all conditions, but underestimates the measured values slightly (< 1 °C) at the very end of a 1 h discharge where the heat generation rate is highest. This is likely a result of the point raised in Section 3.3, where cell X is expected to generate slightly more heat than cell Z at a given time due to the lower temperature of cell X. Since the heat generation rate is highest at the end of discharge, the impacts of this relationship should be most pronounced at the end of discharge. This demonstrates the shortcomings of assuming $\dot{q}_{\mathrm{gen,x}} \approx \dot{q}_{\mathrm{gen,z'}}$ at all times. In this particular case, the high temperatures associated with a 1 h discharge (exceeding 50 °C) indicate that a passively-cooled Tesla Model 3 battery pack should not be subjected to 1 h deep-discharge cycles in commercial applications (though it is feasible with liquid-cooling, see [13]), so this VTS model would be accurate in all practical use cases.

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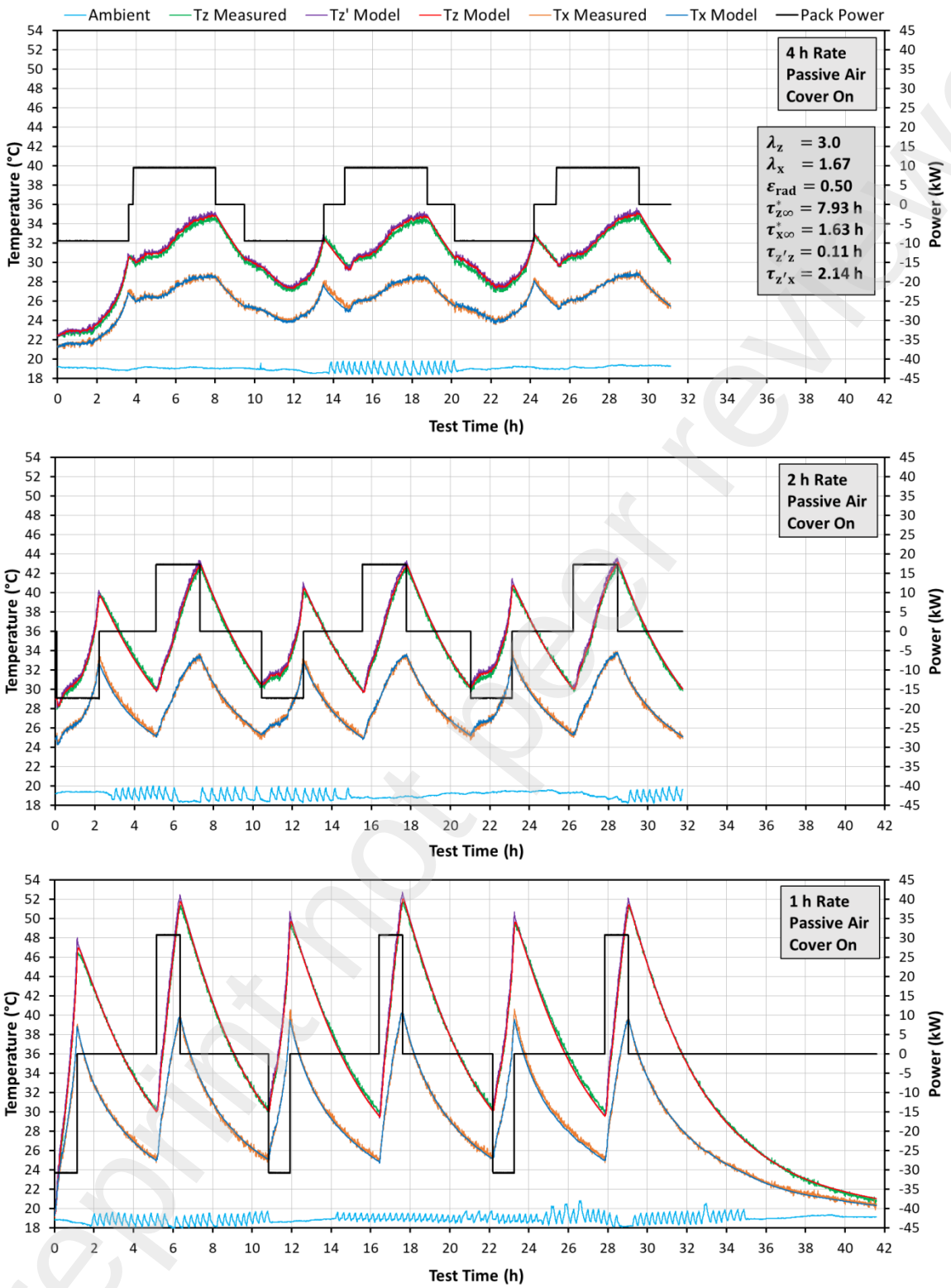


Figure 9 Optimized parameters applied to the VTS model with an empirical dataset featuring three tests at three different constant-power rates.

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The T_z model in Figure 9 is also highly accurate, and the technique of modelling $T_{z'}$ is shown to be effective for reproducing the heat diffusion lag observed in T_z after the end of a 1 h discharge. This is highlighted in Figure 10. After power is interrupted, T_z continues to rise slightly and then tapers off as heat diffusion from the cell interior to the cell surface briefly exceeds the rate of heat transfer away from the cell surface.

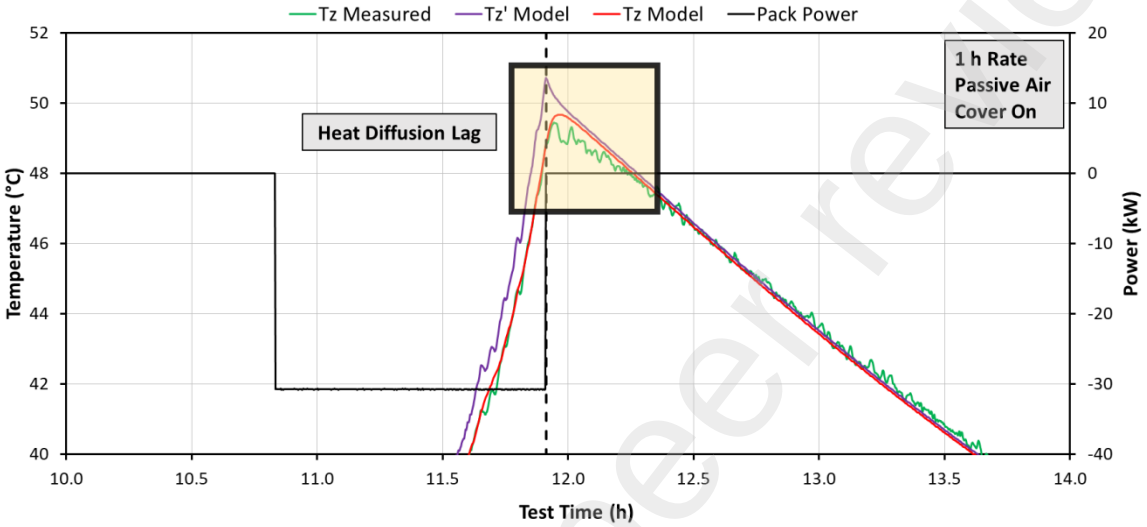


Figure 10 Effectiveness of modelling heat diffusion lag in T_z using a $T_{z'}$ 'pseudo-temperature'.

It should be emphasized here that $T_{z'}$ cannot be interpreted as a true representation of the interior cell temperature. The sensitivity analysis provided in Section S6 of the Supplementary Materials demonstrates how a range of different parameter values can achieve an accurate VTS model with different magnitudes of $T_{z'}$. It is therefore more appropriate to consider $T_{z'}$ as a 'pseudo-temperature' that enables accurate modelling of T_z and T_x but cannot be verified in its own right without direct measurement of the cell interior. Even so, the effectiveness of employing $T_{z'}$ in the thermal model supports the suggestion that the tapered shape of T_z after a 1 h discharge is due to axial heat diffusion from a hotter cell interior.

It is worth mentioning that the complete empirical dataset consisting of the 4 h test, 2 h test, and 1 h test was not necessary for optimizing the parameters in this VTS model. There was no significant difference in the model accuracy at any power rate when only the 1 h test data was used for parameter optimization. However, the model accuracy deteriorated noticeably for the 1 h and 2 h tests when only the 4 h test data was used for parameter optimization. This difference is likely due

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to the fact that the 1 h test data includes both high-power operation and zero-power operation (during lengthy standby cooling periods), thereby covering the complete operational range of the battery. Heat generation dominates natural convection during 1 h discharge and charge, while the long rests exhibit pure natural convection over a wide temperature range. The VTS model should be trained for both these extremes. In contrast, the 4 h test data does not feature high-power operation and is therefore inadequate for parameter optimization unless this is the maximum rate that will be applied in a given second-life application.

While the 1 h test data is sufficient for parameter optimization, optimizing for three different power rates allows for a more rigorous application of the heat generation constraints described in Section 3.3. These constraints will be assessed in Section 4.3.

4.2. Error decay

In practice, an initial guess for an unknown T_x may introduce as much as 10 °C of error into the VTS model. It is therefore important to consider how the VTS model handles errors. Figure 11 demonstrates an example of this, where the same parameters from Figure 9 are applied to the 2 h dataset. The initial guess $T_{x,0}$ at the beginning of the first discharge is set to a value near $T_{\infty,0}^{meas}$, and the same guess is applied at the beginning of the first charge. This shows how T_x converges to T_x^{meas} within 2-3 hours in spite of the significant error introduced. This 'error decay' is plotted directly in the right chart of Figure 11.

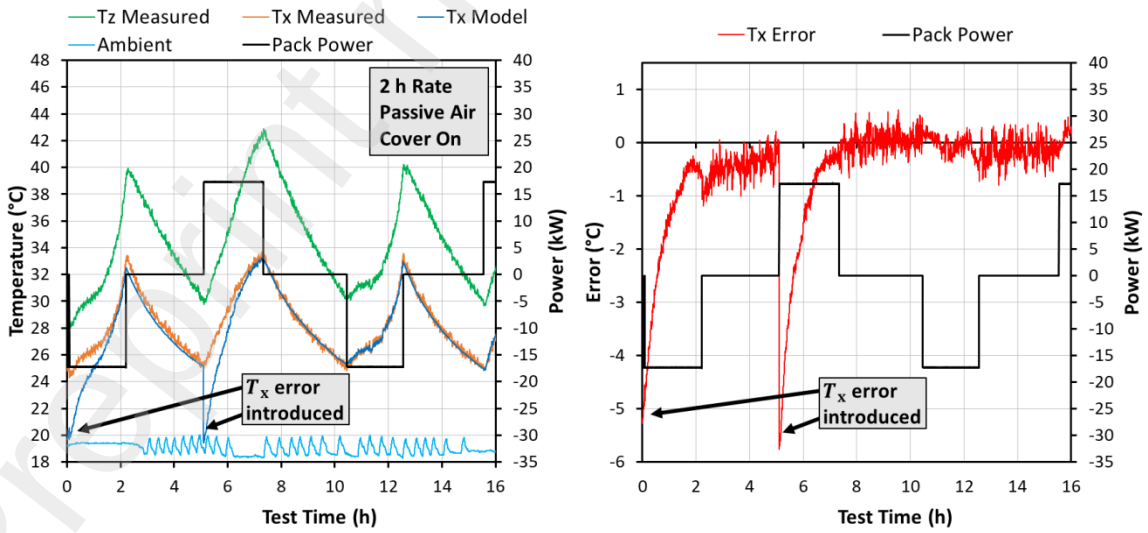


Figure 11 Illustrative example of inherent error decay in the VTS model.

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The inherent error decay of the VTS model can be explained by examining the system equations from Section 2.4. For a given time interval j, the temperature $T_{\mathrm{x},j}$ of cell X can be modelled from a given starting point $T_{\mathrm{x},0}$ using Eq. (58).

$$T_{\mathrm{x},j} = T_{\mathrm{x},0}\mathrm{e}^{-\frac{t_j}{\tau_{\mathrm{x},j}}} + \sum_{i=1}^{j3} \mathrm{e}^{\frac{t_i - t_j}{\tau_{\mathrm{x},i}}} \left(\Theta_{\mathrm{x},i} + \frac{1}{\tau_{\mathrm{x},i}} T_{\infty,i}^{\mathrm{meas}} \right) \Delta t_i \tag{58}$$

The measured value $T_{\mathrm{x},j}^{\mathrm{meas}}$ can be considered to follow the same mathematical formula with an additional term for the measurement noise $T_{\mathrm{x},j}^{\mathrm{noise}}$, as shown in Eq. (59). In this case, the expression depends on a hypothetical time variable $\tau_{\mathrm{x}}^{\mathrm{true}}$ and hypothetical adiabatic temperature rate $\Theta_{\mathrm{x}}^{\mathrm{true}}$. These are the 'true' values of these variables which the VTS model aims to approximate.

$$T_{\mathrm{x},j}^{\mathrm{meas}} = T_{\mathrm{x},0}^{\mathrm{meas}}\mathrm{e}^{-\frac{t_j}{\tau_{\mathrm{x},j}^{\mathrm{true}}}} + \sum_{i=1}^{j} \mathrm{e}^{\frac{t_i - t_j}{\tau_{\mathrm{x},i}^{\mathrm{true}}}} \left(\Theta_{\mathrm{x},i}^{\mathrm{true}} + \frac{1}{\tau_{\mathrm{x},i}^{\mathrm{true}}} T_{\infty,i}^{\mathrm{meas}} \right) \Delta t_i + T_{\mathrm{x},j}^{\mathrm{noise}} \tag{59}$$

If the VTS model is well designed, then $\tau_{\mathrm{x}} \approx \tau_{\mathrm{x}}^{\mathrm{true}}$ and $\Theta_{\mathrm{x}} \approx \Theta_{\mathrm{x}}^{\mathrm{true}}$. This gives Eq. (60) as an expression for the error in T_{x}.

$$T_{\mathrm{x},j} - T_{\mathrm{x},j}^{\mathrm{meas}} = \left(T_{\mathrm{x},0} - T_{\mathrm{x},0}^{\mathrm{meas}}\right)\mathrm{e}^{-\frac{t_j}{\tau_{\mathrm{x},i}}} - T_{\mathrm{x},j}^{\mathrm{noise}} \tag{60}$$

This shows that an instantaneous error in T_{x} (such as an incorrect initial guess) will follow an exponential decay curve dependent on time t and the time variable τ_{x}. Smaller positive values of τ_{x} (indicating faster heat dissipation) will cause the error to decrease at a faster rate. If the assumption of $\Theta_{\mathrm{x}} \approx \Theta_{\mathrm{x}}^{\mathrm{true}}$ breaks down temporarily at the end of a high-power discharge (as was the case in Figure 9), this becomes an instantaneous error in T_{x} which will decay over time as long as the assumption of $\Theta_{\mathrm{x}} \approx \Theta_{\mathrm{x}}^{\mathrm{true}}$ becomes valid again. Standby periods will therefore expedite error decay since $\Theta_{\mathrm{x}} = \Theta_{\mathrm{x}}^{\mathrm{true}} = 0$ during standby. As demonstrated in Figure 11, error decay occurs even if the battery is operational, which is a useful feature for commercial applications.

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4.3. Heat generation

The calculated adiabatic temperature change for cell X and cell Z during each discharge and charge are plotted in Figure 12. The final values of each discharge and charge are sustained during the subsequent standby period (where $\Theta = 0$). These sustained values are annotated in Figure 12 for the third cycle of each test to illustrate how the values of $\Delta T_{\mathrm{ad,x}}^{\mathrm{dis}}$, $\Delta T_{\mathrm{ad,x}}^{\mathrm{chg}}$, $T_{\mathrm{ad,z}}^{\mathrm{dis}}$, and $\Delta T_{\mathrm{ad,z}}^{\mathrm{chg}}$ are determined for each test dataset.

It is interesting to note in Figure 12 that $\Delta T_{\mathrm{gen}}^{\mathrm{chg}} > \Delta T_{\mathrm{gen}}^{\mathrm{dis}}$ for all three tests. This is due to the effects of reversible heating [17]. During discharge, the deintercalation of graphite causes a net increase in cell entropy (endothermic) which subtracts from the irreversible heat generation. During charge, the intercalation of graphite causes a net decrease in cell entropy (exothermic) which adds to the irreversible heat generation. This means that the total heat generation will be greater for a given charge than for a comparable (same current and SOC window) discharge. Since the tests in Figure 12 are constant-power tests, the discharge current will be slightly greater than the charge current, but this does not appear to make a significant difference in Figure 12. Note that the $\Delta T_{\mathrm{gen}}^{\mathrm{chg}} > \Delta T_{\mathrm{gen}}^{\mathrm{dis}}$ difference is less pronounced at higher power rates in Figure 12 since the reversible component of total heat generation is smaller at larger currents.

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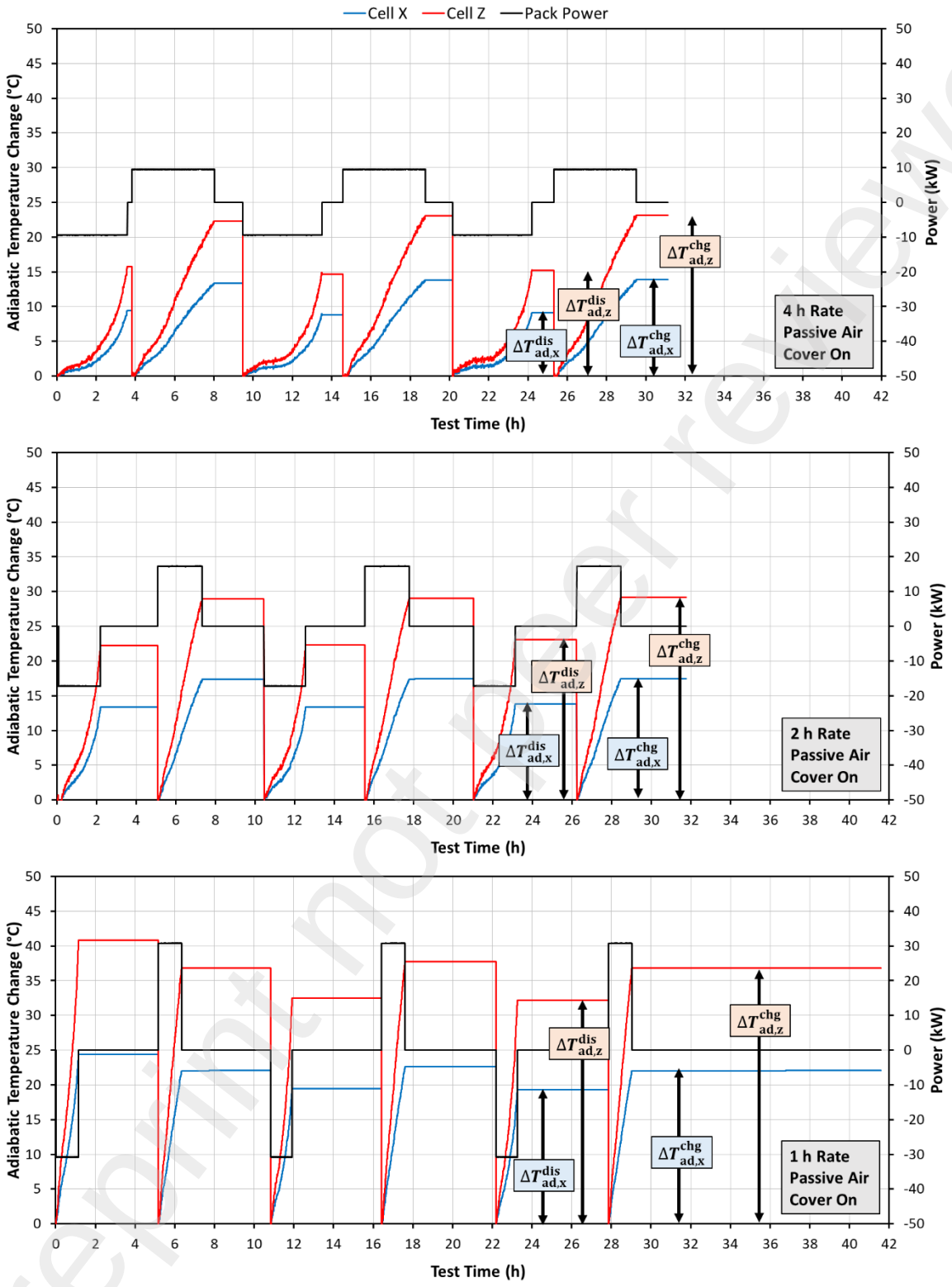


Figure 12 Adiabatic temperature change datapoints used to determine cell heat capacity.

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The calculated adiabatic temperature change values are summarized in Table 2, along with the corresponding $q_{\text{gen}}^{\text{cyc}}$ values (derived in Section 3.3) and calculations of $m_{\text{x}}c_p$ and $m_{\text{z}'}c_p$. Note that the adiabatic temperature change values are larger for cell Z because of the mass ratio $\lambda_{\text{x}} = 1.67$, but the heat generation values are similar for both cells.

Table 2 Heat capacity calculations for each cell and constant-power rate.

	Cell X			**Cell Z (Z' only)**		
Calculation	**4 h Rate**	**2 h Rate**	**1 h Rate**	**4 h Rate**	**2 h Rate**	**1 h Rate**
$q_{\text{gen}}^{\text{cyc}}$ (kJ)	2.25	2.98	4.03	2.24	2.93	3.95
$\Delta T_{\text{ad}}^{\text{dis}}$ (K)	9.1	13.8	19.3	15.2	23.1	32.2
$\Delta T_{\text{ad}}^{\text{chg}}$ (K)	13.9	17.5	22.0	23.3	29.2	36.8
$mc_p\ \left(\frac{\text{J}}{\text{K}}\right)$	98.0	95.2	97.6	58.2	56.1	57.3

The values in Table 2 are used to calculate mean cell heat capacities of $m_{\text{x}}c_p = 96.9\frac{\text{J}}{\text{K}}$ and $m_{\text{z}'}c_p = 57.2\frac{\text{J}}{\text{K}}$ with relative standard deviations of 1.6% and 1.8%, respectively . The fact that the standard deviations are within 2% of the mean values demonstrates that the VTS model has achieved the high level of mc_p consistency that should be expected. The mean values give a ratio of $\frac{m_{\text{x}}}{m_{\text{z}'}} = 1.69$ which agrees well with $\lambda_{\text{x}} = 1.67$ obtained from parameter optimization.

The value of $m_{\text{z}'}c_p$ can be scaled to the full cell Z value using the value of $\lambda_{\text{z}} = 3.0$ obtained from parameter optimization. These results are stated in Eq (61).

$$m_{\text{cell}}c_p = m_{\text{z}'}c_p\left(\frac{\lambda_{\text{z}} + 1}{\lambda_{\text{z}}}\right) = 76.3 \pm 1.8\%\ \frac{\text{J}}{\text{K}} \tag{61}$$

This value is within the 59–78 $\frac{\text{J}}{\text{K}}$ range reported in [14], which gives further merit to this VTS model and the optimized parameter values. The fact that the calculated value is towards the higher end of the reported range may suggest that $m_{\text{cell}}c_p$ includes a small amount of heat capacity from the adjacent liquid channel and filler material.

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4.4. Parameter validation

Table 3 compares the optimized parameter values from Figure 9 to the initial estimates from Section 3.3. To facilitate comparison, the VTS model parameters are converted back to physical parameters by reversing the conversion methods provided in Section S4 of the Supplementary Materials. Note that Table 3 shows just one of many viable sets of parameter values. Section S6 of the Supplementary Materials provides a sensitivity analysis which explores the optimal ranges of each parameter in more detail. In brief, the sensitivity analysis shows that this VTS model has substantial flexibility due to the use of a pseudo-temperature for modelling two-dimensional heat diffusion in cell Z. However, the availability of published experimental values for lithium-ion battery thermal properties makes it possible to better constrain the VTS model within physical expectations. The optimized parameter values in Table 3 are selected to best reflect these physical expectations.

Table 3 Comparison of optimized physical parameters to initial estimates and constraints.

Parameter	Estimated Value	Lower Bound	Upper Bound	Optimized Value
$m_{\text{cell}} c_p \left(\frac{\text{J}}{\text{K}}\right)$	67	59	78	76.3
$\frac{m_{\text{z}'}}{m_{\text{cell}}}$	0.80	0.6	0.97	0.75
$\frac{m_{\text{x}}}{m_{\text{cell}}}$	1.5	1.0	2.0	1.25
ε_{rad}	0.50	0.0	1.0	0.50
$h^*_{\text{conv,z}} \left(\frac{\text{W}}{\text{m}^2\text{K}}\right)$	4.49	0.1	6	0.75
$h^*_{\text{conv,x}} \left(\frac{\text{W}}{\text{m}^2\text{K}}\right)$	5.06	0.1	6	2.52
$k_{\text{z}'\text{z}} \left(\frac{\text{W}}{\text{mK}}\right)$	30	10	50	14.9
$k_{\text{z}'\text{x}} \left(\frac{\text{W}}{\text{mK}}\right)$	1.0	0.1	5	1.89

The optimized value of $\frac{m_{\text{z}'}}{m_{\text{cell}}} = 0.75$ is within the 0.73 – 0.86 range reported in [14], and corresponds to 25% of the cell mass being inactive. The optimized value of $\frac{m_{\text{x}}}{m_{\text{cell}}} = 1.25$ means that cell X is modelled with 25% excess mass compared to cell Z to account for the liquid channel and liquid outlet extending from cell X at the corner of the module. This seems like a reasonable value based on a visual assessment of the battery. It should be emphasized that $\frac{m_{\text{x}}}{m_{\text{cell}}}$ does not necessarily

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represent the true extra heat capacity of the liquid channel and liquid outlet extending from cell X. Rather, $\frac{m_{\mathrm{x}}}{m_{\mathrm{cell}}}$ is a practical method of accounting for the heat diffusion from cell X to these inactive components without actually modelling this heat diffusion, which would require additional temperature sensors and VTS model parameters.

The optimized value of $\varepsilon_{\mathrm{rad}} = 0.50$ is in the middle of the $0.0 \leq \varepsilon_{\mathrm{rad}} \leq 1.0$ emissivity range. The optimized value of $h^*_{\mathrm{conv,z}} = 0.75 \frac{\mathrm{W}}{\mathrm{m^2K}}$ is about 17% of the initial estimate of $4.49 \frac{\mathrm{W}}{\mathrm{m^2K}}$, while the optimized value of $h^*_{\mathrm{conv,x}} = 2.52 \frac{\mathrm{W}}{\mathrm{m^2K}}$ is about 50% of the initial estimate of $5.06 \frac{\mathrm{W}}{\mathrm{m^2K}}$. Both convection coefficients are smaller than their initial estimates, which is expected since the estimates neglect the thermally resistive effects of the pack housing cover. It is interesting to note that $h^*_{\mathrm{conv,z}}$ is substantially smaller than $h^*_{\mathrm{conv,x}}$ despite their similar initial estimates. This may reflect the fact that there is more air space between the module and housing cover at the module corner compared to its top surface. Furthermore, there is additional heat capacity above cell Z and cell X due to the electronics housing at the rear of the pack (see Figure 1 and Figure 3), which further inhibits heat dissipation from cell Z.

The optimized value of $k_{\mathrm{z'x}} = 1.89 \frac{\mathrm{W}}{\mathrm{mK}}$ is 89% higher than the initial estimate of $1.0 \frac{\mathrm{W}}{\mathrm{mK}}$. This may indicate that the unknown properties of the module filler material are beneficial for heat conduction. The optimized value of $k_{\mathrm{z'z}} = 14.9 \frac{\mathrm{W}}{\mathrm{mK}}$ is just below the $20 - 40 \frac{\mathrm{W}}{\mathrm{mK}}$ range reported by Steinhardt et al. [14]. It should be noted that Steinhardt et al. only found ten reported values for axial thermal conductivity $k_{\parallel}$ in their review, and none of these values were for NCA/graphite cells, so it is possible that that the Tesla Model 3 cell has a lower $k_{\parallel}$ than the ten reported values. The result for $k_{\mathrm{z'z}}$ is likely also influenced by the simplifications of the employed thermal model, which separates cell Z into two discrete masses. Even so, both $k_{\mathrm{z'x}}$ and $k_{\mathrm{z'z}}$ give reasonable values, and $k_{\mathrm{z'z}}$ is about an order of magnitude larger than $k_{\mathrm{z'x}}$, which reflects the expected difference between $k_{\parallel}$ and $k_{\perp}$.

Overall, the optimized value of each parameter can be justified in comparison to physical expectations, and this validation gives merit to the VTS model. It should be emphasized again that

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the model parameters serve as *effective* values for a simplified thermal model and should not be interpreted as the true properties of the battery.

4.5. Adaptation to different test conditions

In order to demonstrate the robustness of the VTS model, the seven VTS model parameters were re-optimized for a second empirical dataset. To obtain this second dataset, the 4 h, 2 h, and 1 h constant-power tests described in Section 3.1 were repeated with the battery housing cover removed, since this changes the rate of convection and radiation from the cells. The seven VTS model parameters were initialized to the values obtained from the original parameter optimization (see Figure 9), and only $\tau_{z\infty}$, $\tau_{x\infty}$ and ε_{rad} were permitted to change in the re-optimization. If the VTS model is well-designed, then it should be able to fit this second dataset using the same values of λ_z, λ_x, $\tau_{z'z}$, and $\tau_{z'x}$, since these parameters should be independent of the housing cover. It was found that the VTS model remained insensitive to ε_{rad} despite the change in environmental conditions, so this parameter was fixed at $\varepsilon_{rad} = 0.5$ so that only $\tau_{z\infty}$ and $\tau_{x\infty}$ could be re-optimized.

The results of the re-optimization are shown in Figure 13. Five of the seven parameters have the same values from the covered pack tests, but $\tau_{z\infty}$ has decreased from 7.93 h to 1.65 h and $\tau_{x\infty}$ has decreased from 1.63 h to 1.59 h. The decrease in $\tau_{z\infty}$ and $\tau_{x\infty}$ corresponds to an increased rate of heat dissipation from both cells, which should be expected in the absence of the housing cover. With only these two changes, the VTS model is able to give an accurate fit to the new empirical dataset at all three power rates, and the accuracy is on par with the original dataset. The notable difference between Figure 13 and Figure 9 is that Figure 13 features a larger difference between T_z and $T_{z'}$ during operation. This makes physical sense since cell Z is now able to dissipate more heat directly to the environment, which will result in a larger temperature gradient between the core and the surface. Even so, $T_{z'}$ still quickly converges to T_z once power is interrupted. This shows that the VTS model is robust and adaptable to different environmental conditions.

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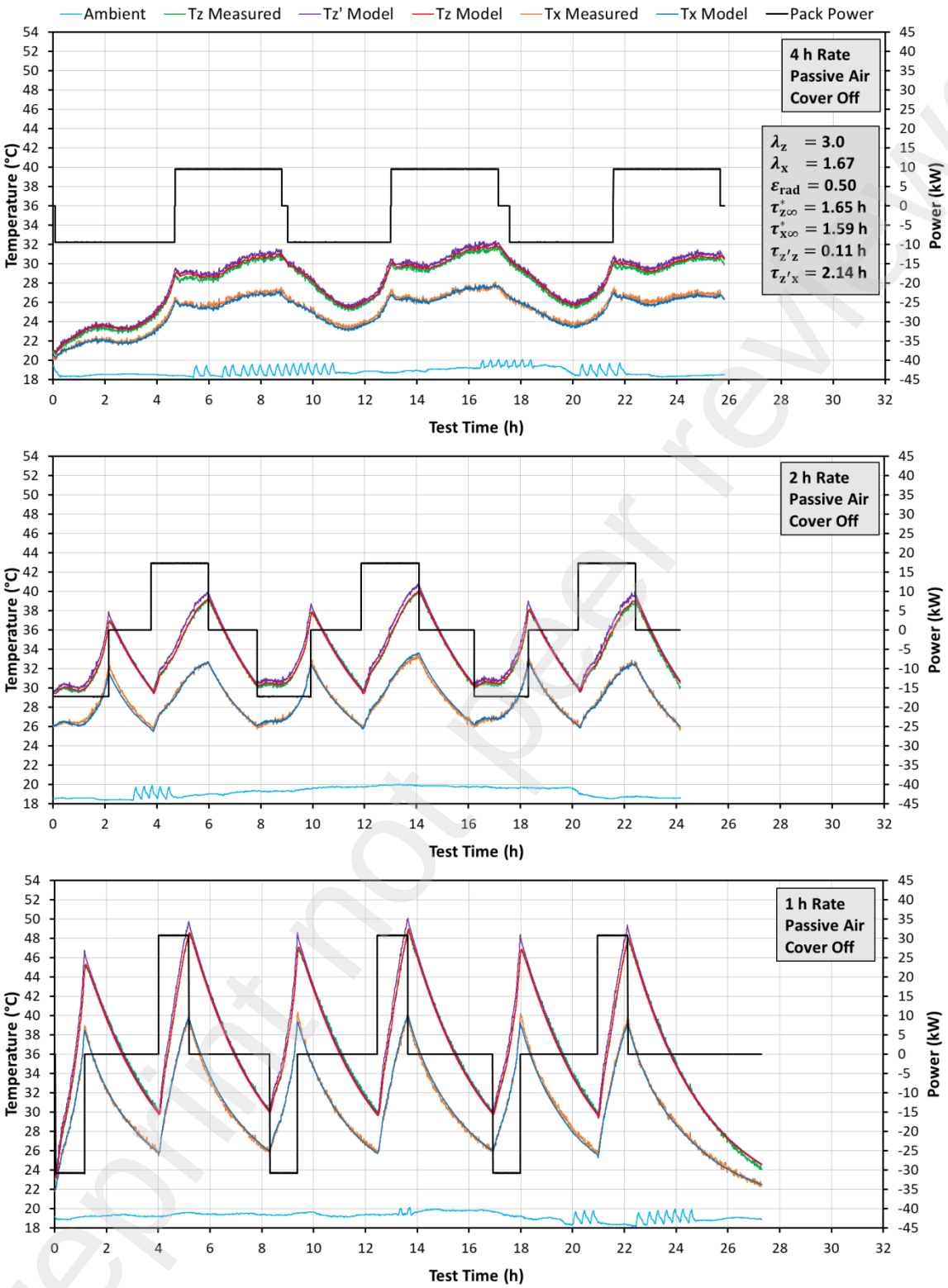


Figure 13 VTS model results with re-optimized values of heat dissipation parameters $\tau_{z\infty}$ and $\tau_{x\infty}$ based on a second empirical dataset with the battery housing cover removed.

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4.6. Practical case study

The empirical datasets presented in Section 4.1 and Section 4.5 are based on T-type thermocouples applied to the inside of a Tesla Model 3 SR+ battery pack after removing the housing cover. In a practical second-life application with an as-received battery pack, the VTS model would be applied to real-time temperature data measured by Tesla's built-in thermistors and read by Tesla's built-in BMS. The VTS model must therefore be adaptable to this data input.

Figure 14 shows an example of thermistor data from a Tesla Model 3 long-range (LR) battery pack, read directly from Tesla's built-in BMS with a 5 s sampling interval. This battery pack was tested as-received without any attempt to remove the housing cover. Passive cooling was employed for these particular tests. The Model 3 LR features longer modules than the Model 3 SR+, so the optimized parameter values determined for a Model 3 SR+ do not necessarily apply to a Model 3 LR. However, these two battery types are quite similar at the rear of the pack where T_{z} and T_{x} are located, so the Model 3 LR dataset provides an opportunity to illustrate how the VTS model can be applied in practice.

The test protocol shown in Figure 14 features a duty-cycle designed to simulate a realistic energy storage power profile for a battery deployed in an electricity grid energy market in the USA. The power is limited to a 4 h rate and the battery SOC is permitted to fluctuate within cell voltage bounds of 3.0 V and 4.15 V. Due to the thermally uninsulated enclosure used for this test, the ambient temperature also fluctuates between 12 °C and 18 °C on a 24 h cycle with the outdoor environment. Both the power profile and ambient conditions therefore differ from the empirical datasets used to optimize the VTS model parameters.

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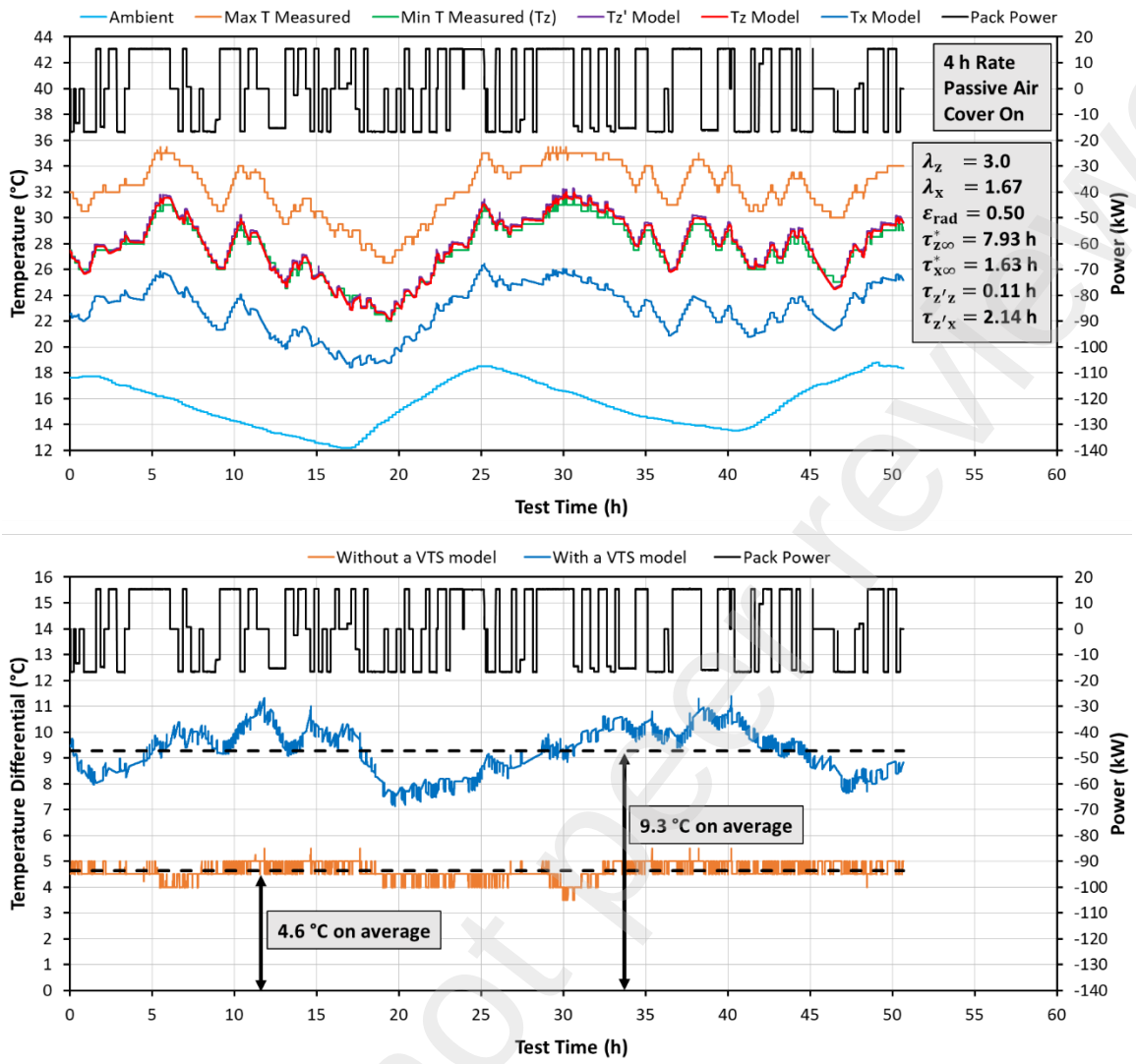


Figure 14 Example of how a VTS model can be used to predict a more accurate battery pack temperature differential during second-life operation based on real-time BMS temperature data and pre-determined VTS model parameters.

The Tesla BMS data reported T_{high} corresponding to a thermistor located in one of the middle modules, and $T_{\text{low}} = T_{\text{z}}$ corresponding to the thermistor attached to cell Z in the outer module. Using the seven optimized parameter values from Figure 9, the VTS model is applied to predict the temperature of cell X at the module corner based on the measured thermistor data from cell Z. The results in Figure 14 show that the VTS model is able to accurately reproduce the T_{z} trend and generate a T_{x} trend using the system equations described in Section 2. Since T_{x} is unknown in this case, the initial guess for $T_{\text{x,0}}$ is taken as the average of $T_{\text{z,0}}$ and $T_{\infty,0}$ with the understanding that

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any error in this guess will dissipate in a few hours as long as the parameter values are appropriate (as discussed in Section 4.2).

Since the Model 3 LR battery pack remains unopened, there is no way to verify the accuracy of the T_x trend shown in Figure 14, and since the parameter values were optimized for Model 3 SR+ battery pack under different test conditions, there is likely an error of 1 – 2 °C in T_x. However, this illustrative example demonstrates how a VTS model can provide important information regarding second-life battery thermal performance that would otherwise be inaccessible. The bottom chart in Figure 14 plots the pack temperature differential calculated from the Tesla thermistor measurements alone (i.e. no VTS model) and the pack temperature differential calculated by the VTS model. Without a VTS model, the system operator would believe the pack temperature differential to be about 4.6 °C, but the VTS model shows that the true pack temperature differential is at least double this value (nearly 10 °C). Battery system operators typically prefer pack temperature differentials below 5 °C to mitigate degradation divergence [18], [19], so this finding is significant. If a second VTS model were developed to predict the temperature near the centre of the battery pack, the true pack temperature differential would likely be revealed to be even higher (more than 10 °C), and this difference would be more pronounced at greater power rates. The improved accuracy of these important thermal performance metrics can be used by system operators in the control of real-time battery operation and the prediction of remaining useful life.

5. Future VTS research

This study has demonstrated a proof-of-concept for VTS technology, but more research is needed before VTS models can be reliably deployed in commercial second-life battery projects. This section will highlight specific recommendations for future work in this field.

5.1. Heat generation modelling

As described in Section 2.5, the VTS model presented in this study assumes that $\dot{q}_{\mathrm{gen,x}} \approx \dot{q}_{\mathrm{gen,z'}}$ at all times. For this particular VTS model, this assumption proved adequate with only a slight limitation when discharging to a low voltage limit at a 1 h power rate. However, it may not always be possible to create an accurate VTS model using this assumption. This point is illustrated in

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Figure 15, which plots the cell voltage and total heat generation during the third cycle of the 4 h, 2 h, and 1 h tests as was done in Section 3.3.

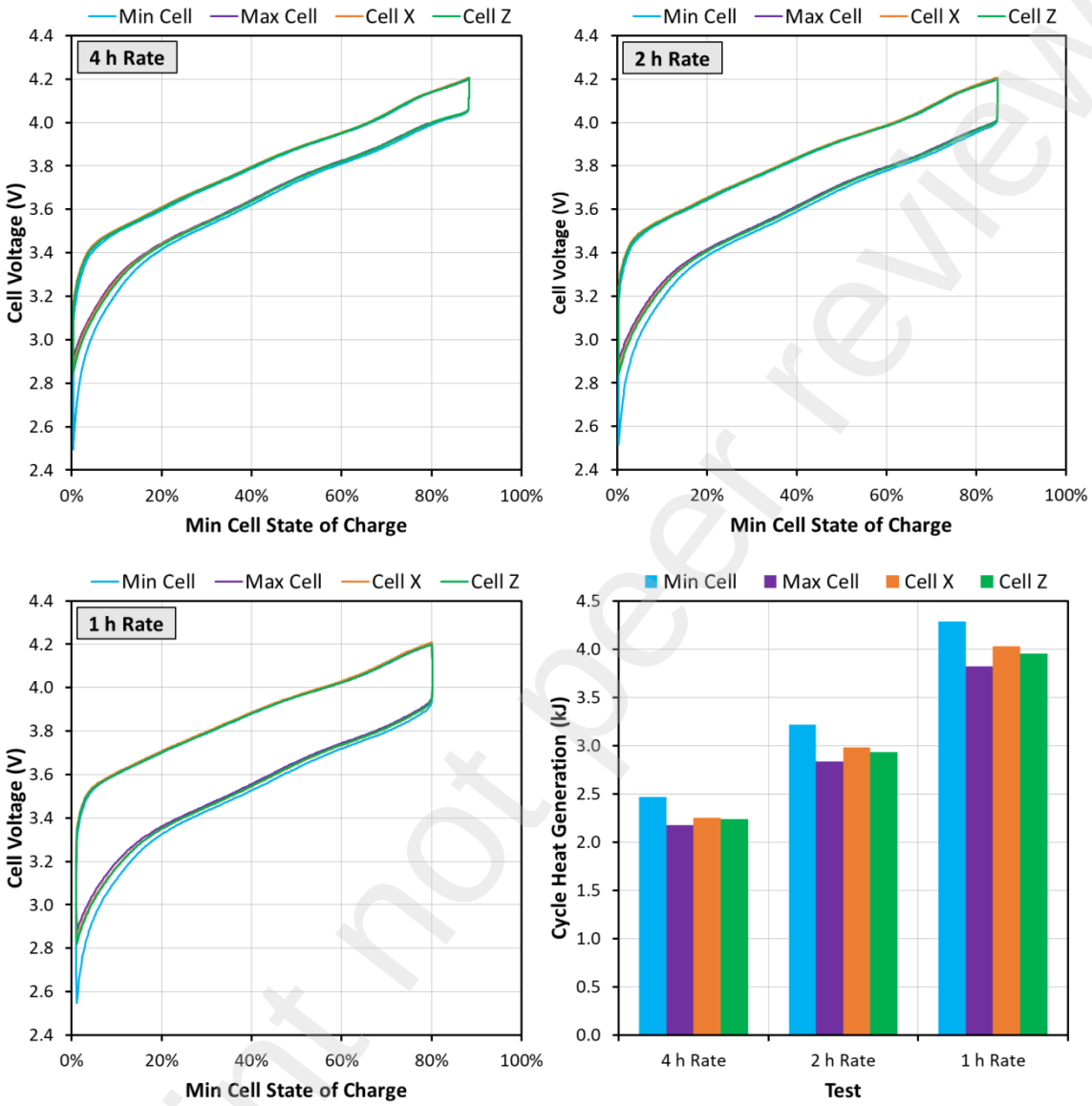


Figure 15 Voltage and heat generation for four different cells in the Tesla Model 3 SR+ battery pack during a single constant-power cycle at three different power rates.

Figure 15 compares the measurements for cell X and cell Z to the two cells which exhibited the lowest and highest voltage during discharge (respectively termed ‘min cell’ and ‘max cell’). This shows that while cell X and cell Z happen to have similar cycle heat generation (within about 2%), the min cell and max cell have a much greater difference (about 13%). If cell X and cell Z had

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exhibited a 13% difference in cycle heat generation, then the $\dot{q}_{\mathrm{gen,x}} \approx \dot{q}_{\mathrm{gen,z'}}$ assumption may not have proved adequate for this VTS model.

The 2019 Tesla Model 3 SR+ battery pack used in this pilot study was at approximately 87% SOH at the time of testing. Since most second-life battery packs will be acquired at 60% – 80% SOH, it is likely that commercial second-life batteries will exhibit an even greater cell-to-cell variance in heat generation than shown in Figure 15, and this variance will likely increase as batteries continue to degrade in second life. This means that a commercially viable VTS model should include a heat generation model which quantifies the relative real-time heat generation rate of one cell or module compared to another. As mentioned in Section 2.5, the goal of such a heat generation model would be to estimate the ratio shown in Eq. (62), where T_{x} and $T_{\mathrm{z'}}$ are absolute temperatures.

$$\frac{\dot{q}_{\mathrm{gen,x}}}{\dot{q}_{\mathrm{gen,z'}}} = \frac{I_{\mathrm{x}}\left(V_{\mathrm{x}} - V_{\mathrm{rev}} + T_{\mathrm{x}}\frac{\partial V_{\mathrm{rev}}}{\partial T}\right)}{I_{\mathrm{z'}}\left(V_{\mathrm{z'}} - V_{\mathrm{rev}} + T_{\mathrm{z'}}\frac{\partial V_{\mathrm{rev}}}{\partial T}\right)} \tag{62}$$

Since all that is needed is a ratio rather than the actual $\dot{q}_{\mathrm{gen}}$ values, precise estimates for V_{rev} and $\frac{\partial V_{\mathrm{rev}}}{\partial T}$ may not be necessary. If these quantities cannot be determined experimentally for a given EV battery pack, it may suffice to make estimates based on published data for similar cell chemistries. However, both V_{rev} and $\frac{\partial V_{\mathrm{rev}}}{\partial T}$ are functions of cell SOC, and different cells in an aged battery pack may have different SOC values at a given time, especially if SOH is non-uniform. This means that cell-specific SOC and SOH estimates may be necessary for an accurate calculation of the heat generation ratio. Many modern EV batteries feature an advanced BMS capable of estimating SOH and SOC for each cell group in the pack [20], so this information should be available for many second-life EV battery packs. If cell-specific SOH and SOC information is not directly available from the built-in BMS, the BMS cell voltage data should be sufficient input for an aftermarket SOC/SOH estimation algorithm, such as those referenced in Chapter 3. Figure 16 illustrates how such an algorithm could be integrated into a more advanced VTS model that compensates for non-uniform battery degradation. Development of this more advanced VTS model is outside the scope of this study, but it is recommended as a worthwhile opportunity for future research.

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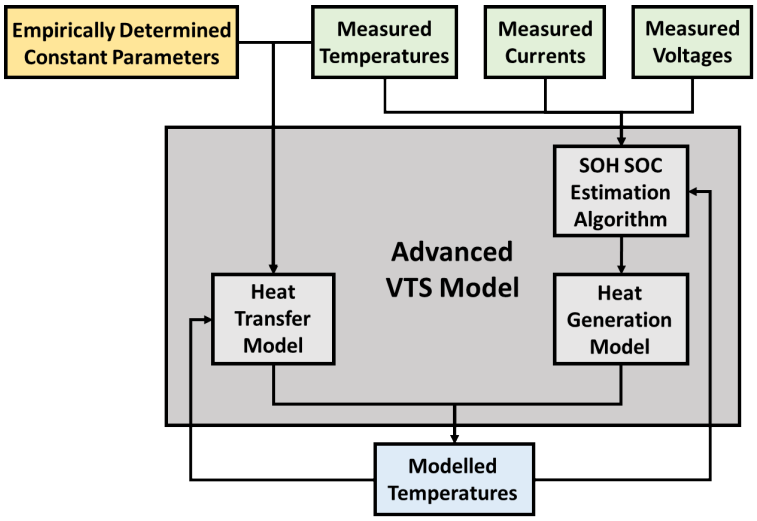


Figure 16 Graphical summary of advanced VTS model concept proposed for future research.

While an advanced VTS model would ensure reliable accuracy in all operational scenarios, the cell voltage curves in Figure 15 suggests that most of the cell-to-cell variance in internal resistance occurs when discharging below 20% SOC. This indicates that the $\dot{q}_{\mathrm{gen,x}} \approx \dot{q}_{\mathrm{gen,z'}}$ assumption may be sufficient for use cases where battery SOC is intentionally maintained above 20%. This operational constraint is sometimes applied in commercial applications to ensure safe and reliable operation [21], [22]. Future VTS research could explore whether the $\dot{q}_{\mathrm{gen,x}} \approx \dot{q}_{\mathrm{gen,z'}}$ assumption is sufficient in specific SOC ranges for aged second-life batteries. When this assumption can be applied, the VTS model requires no knowledge of cell SOC or SOH.

5.2. Diverse training data

The VTS model presented in this study was trained using an empirical dataset consisting of constant-power tests at three different power rates in a relatively consistent ambient environment. While this training data was sufficient for a proof-of-concept, it may not adequately capture the full range of operational scenarios and ambient conditions that arise in commercial second-life energy storage projects. Figure 14 already demonstrated how commercial projects can feature a fluctuating power profile and a varying ambient temperature. To ensure VTS reliability in such conditions, the VTS model parameters should be optimized using training data that captures this variability. Future VTS research should explore a wider range of training data and verify VTS reliability in a wide range of conditions. Suggested conditions include cold and warm environments, fluctuating environments, frequency regulation duty cycles, and stacked-service

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duty-cycles. The results of this pilot study suggest that a single high-power deep-discharge test with lengthy standby cooling periods may be sufficient training data to optimize VTS model parameters for any operational scenario, but this must be explored in future research.

One notable consideration is the handling of radiation in the heat dissipation parameters. The methods described in Section S3 of the Supplementary Materials include a linearization of the radiation equation, and this linearization is only reliable for a given ambient temperature T_∞. If the VTS model is deployed in an environment where T_∞ varies significantly (e.g. 10 °C daily fluctuation), the linearization would introduce an inaccuracy into the VTS model. New methods could be developed to build T_∞ compensation into the heat dissipation parameters. However, the VTS model presented in this study proved to be fairly independent of $\varepsilon_{\mathrm{rad}}$, so such methods may not be necessary. Future research should explore whether $\varepsilon_{\mathrm{rad}}$ is more impactful when the VTS model is applied at different T_∞ values (e.g. 15 °C vs 30 °C).

5.3. Customized parameter optimization

This study employed a common parameter optimization software in order to demonstrate a proof-of-concept for VTS models, but it may be beneficial to develop parameter optimization algorithms that are tailored to VTS models. This may be especially necessary for more complex VTS models with many unknown parameters. Parameter optimization algorithms are outside the scope of this study, but it is recommended as a consideration for future VTS research.

5.4. Other VTS models

This study has presented one VTS model to serve as a proof-of-concept for VTS technology, but many other VTS models could be developed in future research. Each unique EV battery pack design will require a unique VTS model tailored to the architecture and geometry of the given pack. A range of VTS models may also be required for a single battery pack if temperature models are required at multiple locations with different geometries. For example, a VTS model could be developed to model the temperature in the centre of the Tesla Model 3 battery pack based on the measured T_z. This would require a different VTS model design compared to the one presented in this chapter, since a cell in the centre of the pack must be modelled differently than cell X at the module corner (e.g. axial heat diffusion in both cells instead of just one).

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Different thermal management systems will also require different VTS models. For example, the VTS model presented in this study would need to be modified if the Tesla Model 3 SR+ battery pack was to be operated with liquid cooling. In this case, the model would need to account for forced convection between the liquid coolant and the liquid channel. Heat conduction from the cells to the liquid channels would likely dominate the heat conduction within the modules and the natural convection from the modules to the ambient air, so parameters like $\tau^*_{x\infty}$, $\tau^*_{z\infty}$, and $\tau_{z'x}$ may not be necessary in a liquid-cooled VTS model.

Hybrid VTS models may be required for second-life battery projects featuring variable thermal management systems, such that different sets of VTS parameters are activated under different conditions. For example, if a cooling fan is engaged intermittently, then different values of $\tau^*_{x\infty}$ and $\tau^*_{z\infty}$ would need to be activated to account for the switching between natural convection and forced convection. If liquid cooling is engaged intermittently, then two completely different VTS models would need to be activated intermittently to account for the significant change in thermal management. In such cases as these, the VTS model parameters would have to be determined using training data that reflects the operational conditions as closely as possible. Thermal management setpoints such as liquid pressure and flow rate could be used as real-time inputs for the VTS model.

Since there are numerous unique EV battery pack designs, VTS models will need to be tailored specifically to the battery pack in question. While the VTS design process may vary considerably for different EV battery packs, a generalized approach is provided below to guide future research and development:

- A 'pilot pack' for a given EV battery design must be opened and retrofitted with temperature sensors at the desired locations. Care should be taken to ensure that the applied temperature sensors do not impact the thermal properties of the local area.
- A thermal model must be developed to establish relationships between desired temperature locations. This requires visual inspection of the opened pilot pack.
    - Significant simplifications may be necessary in order to keep the number of unknown parameters to a minimum. This study has presented examples of such simplifications, such as employing the mass ratio λ_x to account for heat diffusion away from cell X.

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- The thermal model may need to account for heat diffusion from the cell interior to the temperature sensor location. This study has presented a simple means of accomplishing this using an unverifiable 'pseudo-temperature' to model the cell interior temperature.

- A set of system equations must be developed based on the established thermal model. These equations should derive from a fundamental energy balance for each temperature location in the thermal model, including any pseudo-temperatures.
  - This study has presented novel methods to simplify the system equations into a form which can be solved analytically and adapted numerically. This approach is recommended for future VTS research.
  - The system equations will require ratios to describe the heat generation rates at retrofitted sensor locations compared to built-in sensor locations. Depending on the age and intended application of the battery, it may suffice to assume a unity ratio. Otherwise, a heat generation model may need to be developed which receives real-time BMS cell voltage measurements as inputs. This may also require an algorithm to estimate cell SOH and SOC.
  - When natural convection is the dominant heat dissipation pathway, the heat dissipation parameters will be dependent on temperature. Section S3 of the Supplementary Materials provide a means of accounting for this temperature dependence using constant parameters.
  - The heat dissipation parameters may need to account for radiation in addition to convection. Section S3 of the Supplementary Materials provide a simple means of accomplishing this using a linearized radiation equation and a novel expression for combined convection and radiation.

- A constant value must be determined empirically for each unknown parameter in the VTS model. This requires empirical training data acquired from retrofitted temperature sensors applied to the pilot pack during operational tests.
  - The pilot pack should be restored to its as-received form as closely as possible (other than the retrofitted temperature sensors) for these tests to ensure that its heat

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transfer properties accurately reflect a commercially deployed battery pack of the same design.

- The tests should cover the complete range of power profiles, ambient temperatures, and thermal management conditions that are expected in commercial applications. Data sampling at 10 s intervals is recommended.
- An appropriate parameter optimization software must be selected based on the number of VTS model parameters being optimized. In this study, Microsoft Excel Solver proved sufficient. For more complex models, it may be necessary to develop a customized parameter optimization algorithm.
- Physical parameters should be estimated for the pilot pack as a means of constraining the VTS model parameters during the optimization process. This study also presented a novel method for calculating mc_p values from the VTS model results, which can provide an additional constraint.

- A commercially deployed VTS model must be provided with real-time measurements of battery temperature and ambient temperature at an appropriate sampling rate. Data sampling intervals of 10 s to 60 s are recommended.

6. Conclusions

This study has proposed the concept of virtual temperature sensors as a novel battery monitoring technology. While VTS technology could theoretically be used in any battery application to reduce the number of physical sensors required to monitor battery temperature distributions, this pilot study has focused on implications for second-life battery packs. When used EV battery packs are repurposed as-received to save time and costs, it can introduce significant uncertainty regarding battery thermal performance, especially when deviating from the original thermal management system. VTS technology can mitigate this uncertainty by providing a non-invasive means of monitoring temperatures at any desired location in the battery pack.

A detailed VTS proof-of-concept has been demonstrated for a Tesla Model 3 SR+ battery pack using a simplified thermal model with seven empirically-determined constant parameters. The seven parameters are governed by the physical characteristics of the battery and its surroundings,

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which makes the model versatile in different operating conditions. The proposed numerical methods are simple enough to be executed in a standard Microsoft Excel spreadsheet, yet yield highly accurate and robust temperature predictions after parameter optimization. The VTS model has also been shown to exhibit inherent error decay, which enhances the reliability of the predicted temperatures. A preliminary case study showed that VTS technology can provide valuable insights into battery pack temperature distributions that would otherwise go undetected.

This pilot study has also identified a number of opportunities for future VTS research and development. In commercial applications, integrating VTS models with cell-specific SOC and SOH estimation algorithms will likely yield the most reliable results. A generalized approach has also been provided to help guide other researchers in developing VTS models for EV battery packs.

Acknowledgements

This work was made possible with funding from the Natural Sciences and Engineering Research Council of Canada (NSERC, #NETGP 468468-14) through support of the Strategic Network of Energy Storage Technology (NESTnet) based at Toronto Metropolitan University (formerly Ryerson University). Additional support was provided by the Vanier-Banting Secretariat and the Province of Nova Scotia through Graduate Scholarships. Research equipment support was provided by the Canada Foundation for Innovation (CFI #32025), the Nova Scotia Research Innovation Trust (NSRIT, #32025) and the Tyler Lewis Clean Energy Research Foundation.

References

[1] K. Smith and C.-Y. Wang, “Power and thermal characterization of a lithium-ion battery pack for hybrid-electric vehicles,” *J. Power Sources*, vol. 160, no. 1, pp. 662–673, Sep. 2006, doi: 10.1016/j.jpowsour.2006.01.038.

[2] Y. Ye, Y. Shi, N. Cai, J. Lee, and X. He, “Electro-thermal modeling and experimental validation for lithium ion battery,” *J. Power Sources*, vol. 199, pp. 227–238, Feb. 2012, doi: 10.1016/j.jpowsour.2011.10.027.

[3] S. C. Chen, C. C. Wan, and Y. Y. Wang, “Thermal analysis of lithium-ion batteries,” *J. Power Sources*, vol. 140, no. 1, pp. 111–124, Jan. 2005, doi: 10.1016/j.jpowsour.2004.05.064.

[4] Z. Zhang, L. Jia, N. Zhao, and L. Yang, “Thermal modeling and cooling analysis of high-power lithium ion cells,” *J. Therm. Sci.*, vol. 20, no. 6, pp. 570–575, Dec. 2011, doi: 10.1007/s11630-011-0512-3.

[5] K. Yoo and J. Kim, “Thermal behavior of full-scale battery pack based on comprehensive heat-generation model,” *J. Power Sources*, vol. 433, p. 226715, Sep. 2019, doi: 10.1016/j.jpowsour.2019.226715.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4851132

[6] V. V. Ganesan and A. Jain, “Computationally-efficient thermal simulations of large Li-ion battery packs using submodeling technique,” *Int. J. Heat Mass Transf.*, vol. 165, p. 120616, Feb. 2021, doi: 10.1016/j.ijheatmasstransfer.2020.120616.
[7] C. Zhu, X. Li, L. Song, and L. Xiang, “Development of a theoretically based thermal model for lithium ion battery pack,” *J. Power Sources*, vol. 223, pp. 155–164, Feb. 2013, doi: 10.1016/j.jpowsour.2012.09.035.
[8] P. Amiribavandpour, W. Shen, D. Mu, and A. Kapoor, “An improved theoretical electrochemical-thermal modelling of lithium-ion battery packs in electric vehicles,” *J. Power Sources*, vol. 284, pp. 328–338, Jun. 2015, doi: 10.1016/j.jpowsour.2015.03.022.
[9] T. M. Bandhauer, S. Garimella, and T. F. Fuller, “A Critical Review of Thermal Issues in Lithium-Ion Batteries,” *J. Electrochem. Soc.*, vol. 158, no. 3, p. R1, 2011, doi: 10.1149/1.3515880.
[10] S. J. Bazinski and X. Wang, “Predicting heat generation in a lithium-ion pouch cell through thermography and the lumped capacitance model,” *J. Power Sources*, vol. 305, pp. 97–105, Feb. 2016, doi: 10.1016/j.jpowsour.2015.11.083.
[11] L. Ramotar, G. L. Rohrauer, R. Filion, and K. MacDonald, “Experimental verification of a thermal equivalent circuit dynamic model on an extended range electric vehicle battery pack,” *J. Power Sources*, vol. 343, pp. 383–394, Mar. 2017, doi: 10.1016/j.jpowsour.2017.01.040.
[12] R. Schmuch, R. Wagner, G. Hörpel, T. Placke, and M. Winter, “Performance and cost of materials for lithium-based rechargeable automotive batteries,” *Nat. Energy*, vol. 3, no. 4, pp. 267–278, Apr. 2018, doi: 10.1038/s41560-018-0107-2.
[13] C. White and L. G. Swan, “Pack-level performance of electric vehicle batteries in second-life electricity grid energy services,” *J. Energy Storage*, vol. 57, p. 106265, Jan. 2023, doi: 10.1016/j.est.2022.106265.
[14] M. Steinhardt, J. V. Barreras, H. Ruan, B. Wu, G. J. Offer, and A. Jossen, “Meta-analysis of experimental results for heat capacity and thermal conductivity in lithium-ion batteries: A critical review,” *J. Power Sources*, vol. 522, p. 230829, Feb. 2022, doi: 10.1016/j.jpowsour.2021.230829.
[15] T. L. Bergman, *Fundamentals of Heat and Mass Transfer*. John Wiley & Sons, 2011.
[16] “Solver.” Frontline Systems, Inc, Incline Village, NV. [Online]. Available: https://www.solver.com/
[17] H. Yang and J. Prakash, “Determination of the Reversible and Irreversible Heats of a LiNi0.8Co0.15Al0.05O2ÕNatural Graphite Cell Using Electrochemical-Calorimetric Technique,” *J. Electrochem. Soc.*.
[18] A. A. Pesaran, “Battery thermal models for hybrid vehicle simulations,” *J. Power Sources*, vol. 110, no. 2, pp. 377–382, Aug. 2002, doi: 10.1016/S0378-7753(02)00200-8.
[19] Y. Li, Z. Zhou, and W.-T. Wu, “Three-dimensional thermal modeling of Li-ion battery cell and 50 V Li-ion battery pack cooled by mini-channel cold plate,” *Appl. Therm. Eng.*, vol. 147, pp. 829–840, Jan. 2019, doi: 10.1016/j.applthermaleng.2018.11.009.
[20] M. R. M. Kassim, W. A. W. Jamil, and R. M. Sabri, “State-of-Charge (SOC) and State-of-Health (SOH) Estimation Methods in Battery Management Systems for Electric Vehicles,” in *2021 IEEE International Conference on Computing (ICOCO)*, Kuala Lumpur, Malaysia: IEEE, Nov. 2021, pp. 91–96. doi: 10.1109/ICOCO53166.2021.9673580.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4851132

[21] D. M. Davies *et al.*, "Combined economic and technological evaluation of battery energy storage for grid applications," *Nat. Energy*, vol. 4, no. 1, pp. 42–50, Dec. 2018, doi: 10.1038/s41560-018-0290-1.
[22] M. Faisal, M. A. Hannan, P. J. Ker, M. S. Hossain Lipu, and M. N. Uddin, "Fuzzy-Based Charging–Discharging Controller for Lithium-Ion Battery in Microgrid Applications," *IEEE Trans. Ind. Appl.*, vol. 57, no. 4, pp. 4187–4195, Jul. 2021, doi: 10.1109/TIA.2021.3072875.

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